A Survey on the Development of Human-Centered Artificial Agents for Fighting Games

Martinho Tavares, João M. Carvalho, and Armando J. Pinho, *Member, IEEE*

(Survey Paper)

***Abstract*—Artificial intelligence methods can be used to develop artificial agents for competitive video games. These agents have the potential to be learning tools for players, as well as provide automated insights into how balanced the game is. This motivates the existence of agents whose skill is adjusted to the player. However, this also assumes the agents are good substitutes for human players. Developing "human-centered" agents, focused on the gameplay experience of human players, is a difficult problem. In the genre of fighting games, they need to account for different characteristics such as human-like adaptation and reaction time, which is an important factor in these games' design. We survey the different dimensions by which this problem is addressed, both in commercial games and the literature. These dimensions include the performance of the agents, their human likeness and the diversity of their behaviors. We thus present the state of the art in the creation of human-centered agents, along with future directions, in which this problem can be explored further.**

***Index Terms*—Artificial intelligence (AI), fighting game, human-like behavior, machine learning (ML).**

I. Introduction

FIGHTING games are competitive two-player zero-sum video games. These are real-time games where both players act simultaneously, as opposed to turn-based play. These games are usually played between human players, but artificial opponents are also implemented. When used as a surrogate for human players, artificial opponents can be beneficial. For one, we can set up controlled learning environments in which new players can learn to play against opponents adjusted to their skill level, facilitating learnability, which is difficult to attain in these games [1]. Another benefit is for playtesting: added to the use of human testers, we can have a population of artificial agents learn to play the game in order to assess its balance [2], [3]. This is especially important considering these games tend to be updated over time, with changes in mechanics that alter the game's balance.

However, many approaches to creating these artificial opponents are not appropriate, in the sense that the opponents do not play like humans [4]. For instance, one major factor in this discrepancy is reaction time: many of the moves performed by each player cannot be reacted to or is difficult to do so, because of this, strategies in fighting games are heavily based on predictions of the opponent's future moves [5]. We can think of these games as multiple consecutive trials of *Rock–Paper–Scissors* to some extent [6], [7]. Fighting games require the player to be as unpredictable as possible while predicting the opponent's behavior patterns, similar to *Rock–Paper–Scissors* [8].

This dynamic however is either not captured by commercial games or developers are not transparent about whether it is present. Transparency in how artificial opponents work is important as players show bias when playing against them, and assume these opponents will not play intelligently or follow reasonable patterns [9]. Another factor we need to consider is the believability of the artificial opponents' behavior. Opponents exhibiting human-like behavior make for more engaging play [10], since they mimic the way people think and so players are more inclined to be competitive against them as they would be against a human opponent.

This motivates the creation of agents that emulate the experience of playing with a human player. Throughout this work, we refer to these agents as "human-centered" agents, deriving from the term of "human-centered design." A human-centered agent is built with a focus on the experience of the end user, taking into account aspects of human play. In this work, we are focused on the following desirable qualities.

1) Performance: The competence of the agent, i.e., its skill level, so that it can appropriately match against more experienced players [11].
2) Adaptability: Ability to adapt quickly to different opponent strategies, as players can detect exploitable patterns in the agent's behavior [12], [13], [14], [15], [16], [17].
3) Human likeness: How credible the agent's behavior is, such that it can be perceived to have qualities of a human player, crucial for player expectations [9], [18], [19], [20].

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面向格斗游戏的以人为中心的人工智能体开发综述

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（综述论文）

摘要——人工智能方法可用于开发竞技电子游戏中的人工智能体。这些智能体有望成为玩家的学习工具，并能自动分析游戏的平衡性。因此，有必要开发能够根据玩家水平调整自身技能的智能体。然而，这也以智能体能够有效替代人类玩家为前提。开发专注于人类玩家游戏体验的“以人为中心的”智能体是一项难题。在格斗游戏中，这类智能体需要考虑类人适应、反应时间等不同特征，其中反应时间是此类游戏设计中的重要因素。本文从多个维度综述了商业游戏和相关文献中应对这一问题的方法。这些维度包括智能体的性能、类人性及其行为的多样性。由此，本文介绍了以人为中心的智能体构建领域的最新进展，以及可进一步探索这一问题的未来方向。

索引词——人工智能（AI）、格斗游戏、类人行为、机器学习（ML）。

I. 引言

FIGHTING游戏是具有竞争性的双人零和电子游戏。这类游戏属于实时游戏，双方玩家同时行动，而非采用回合制玩法。这类游戏通常由人类玩家对战，但也实现了人工对手。用作人类玩家的替代者时，人工对手能够带来诸多益处。首先，我们可以构建受控学习环境，让新玩家在其中与根据其技能水平调整的对手对战，从而提高这类游戏通常难以实现的易学性 [1]。另一个益处体现在游戏测试方面：除了使用人类测试人员之外，我们还可以让一个人工智能体种群学习如何玩这款游戏，以评估其平衡性 [2], [3]。考虑到这类游戏往往会持续更新，而机制的变化会改变游戏平衡性，这一点尤为重要。

然而，许多创建这些人工对手的方法并不合适，因为这些对手的打法并不像人类 [4]。例如，造成这种差异的一个主要因素是反应时间：玩家使出的许多招式让对方来不及反应，或很难做出反应，因此，格斗游戏中的策略在很大程度上依赖于对对手后续招式的预测 [5]。在某种程度上，我们可以将这些游戏视为连续进行的多轮石头剪刀布游戏 [6], [7]。格斗游戏要求玩家在预测对手行为模式的同时，尽可能让自己的行为难以预测，这与石头剪刀布 [8]类似。

然而，商业游戏要么未能体现这种动态博弈，要么开发者并未明确说明游戏中是否存在这种机制。人工对手的运作方式需要透明，因为玩家在与其对战时会表现出偏置，认为这些对手不会采取智能的打法，也不会遵循合理的行为模式 [9]。我们需要考虑的另一个因素是人工对手行为的可信度。表现出类人行为的对手能让对战更具吸引力 [10], ，因为它们模仿了人类的思维方式，因此玩家更愿意像面对人类对手一样，与它们展开竞技。

这推动了能够模拟与人类玩家对战经验的智能体的创建。在本文中，我们将这些智能体称为“以人为中心的”智能体，这一称谓源于“以人为中心的设计”一词。以人为中心的智能体在构建时注重最终用户的经验，并将人类对战的各个方面纳入考量。本文重点关注以下理想特性。

1) 性能：智能体的能力，即其技能水平，使其能够与经验更丰富的玩家进行适当的匹配 [11]。2) 适应性：快速适应不同对手策略的能力，因为玩家能够识别智能体行为中可利用的模式 [12], [13], [14], [15], [16], [17]。3) 类人性：智能体行为的可信程度，使其能够被认为具有人类玩家的特质，这对于满足玩家期望至关重要 [9], [18], [19], [20]。

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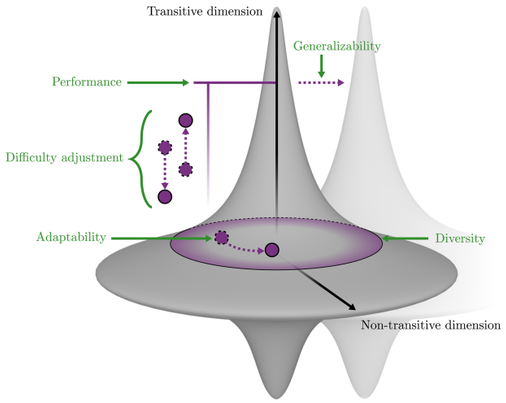


Fig. 1. Influence of human-centered qualities on the strategy space covered by artificial agents. Illustrated using the strategy space geometry of real world games proposed in [31], which resembles a spinning top. Each circle represents an agent following a particular strategy. Human likeness is not represented since it is difficult to illustrate, but its effect is a restriction on which points in the space are valid: those that represent human-like behavior.

4) Difficulty adjustment: Artificial agents can be beneficial to players of various skill levels, and the game should be ready to offer agents appropriate to those skill levels [11], [21], [22], [23].
5) Diversity: A population of agents should cover a range of strategies, which is both beneficial for training [24], [25], [26] and evaluation [22], [24], [27] of artificial agents, as well as providing a wider coverage of the game's strategy space [11], [28].
6) Generalizability: Ability to generalize between different playable characters, in order to economize training costs when building agents spanning a game's character roster [5], [21], generalize between opponent characters [29] and generalize between games [30].

These qualities are the result of grouping the various research goals of the works found in the literature. These are all qualities that are desirable in an artificial agent, but not all are strictly mandatory. We also argue the qualities are not equally important. Human likeness is the most important feature to strive for since without it the gameplay experience will be distant from the one had with human players, regardless of whether the agent exhibits the other qualities.

A human-centered agent should represent human players. Therefore, it should cover the space of human-like strategies. Fig. 1 presents the effect each quality has on the strategy space covered by artificial agents. The strategy space geometry is based on the one proposed in [31], and is explained further in Section VI-E. Human likeness is not illustrated, but it restricts the space to the strategies a human player would likely employ. Performance and diversity help increase the range of players effectively covered by the agent; performance is required for higher skilled players and diversity is required to challenge different playstyles. These are concerned with the space covered in the transitive and nontransitive dimensions, respectively. On the other hand, difficulty adjustment promotes adjustment of the skill level of the agent, while adaptability promotes modification of the agent's strategy in response to the player. These two qualities are concerned with traversing the space in the transitive and nontransitive dimensions, respectively. Finally, generalizability is desirable from a technical perspective, as a general agent can be reused between character matchups, avoiding the combinatorial explosion when tackling all possible matchups [21], and applied easily to games containing different mechanics. It is represented as a different strategy space, which can correspond to a different character matchup or a different game altogether. These are all distinct qualities that may be considered when building an agent for human players, although fulfilling all of them may be difficult.

There exist various approaches to the development of artificial agents, with the most basic being artificial intelligence (AI) methods involving the manual creation of rules that determine the agent's behavior. However, these methods require extensive expert knowledge and labor. Another approach is to use machine learning (ML) methods, which involve autonomous learning of behaviors or rules to accomplish a given task. It is also possible to use evolutionary and planning algorithms, which can be simpler to adapt to the domain of fighting games since they are generally less dependent on evaluating the current game state, and are thus more problem-agnostic. They may be incorporated into other algorithms, which is often done to improve performance. For example, we can mix long-term planning with more reactive algorithms, such as reinforcement learning (RL) [32], [33], or with evolutionary algorithms [34], [35], or simply use evolutionary algorithms with artificial neural networks in general [22], [36], [37], [38].

We perform a survey on existing techniques for development of artificial opponents in fighting games, detailing what they achieve in terms of the qualities of human-centered agents. The rest of this article is organized as follows. Section II introduces the fighting game genre in more detail. Section III details the methodology used for this survey. Sections IV and V introduce existing commercial games and learning environments, respectively, detailing their advancements in the development of artificial agents for fighting games. Sections VI-A to VI-F discuss works within each quality of a human-centered agent listed earlier. Section VII presents summary notes on the contributions each method has on achieving the qualities of a human-centered agent. Finally, Section VIII concludes this article and provides directions for future research.

II. Background

Fighting games are competitive video games in which players fight each other in an arena [39]. Famous examples of these games are those in the *Street Fighter* series. Players have a set amount of health, and the objective is to deplete the opponent's health within the time limit or remain with the health lead otherwise. Fighting games have a multitude of characters, and each player controls one of said characters. Each character has a specific set of offensive and defensive moves that the player has to use to succeed. Fig. 3 presents an example of a game state

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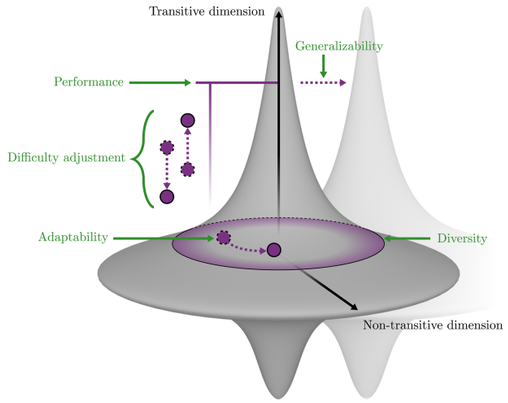


图1. 以人为中心的特质对人工智能体所覆盖的策略空间的影响。图示采用了[31]，中提出的现实世界游戏的策略空间几何结构，其形状类似陀螺。每个圆圈代表一个遵循特定策略的智能体。由于类人性难以图示，因此未予呈现，但其作用是限制空间中哪些点是有效的：只有代表类人行为的点才有效。

4) 难度调整：人工智能体可以帮助不同技能水平的玩家，因此游戏应当能够提供与这些技能水平相匹配的智能体[11]，[21]，[22]，[23]。5) 多样性：智能体群体应涵盖一系列策略，这既有利于人工智能体的训练[24]，[25]，[26]和评估，也能更广泛地覆盖游戏的策略空间[11]，[28]。6) 泛化能力：在不同可玩角色之间进行泛化的能力，以便在构建覆盖游戏全部角色的智能体时节约训练成本[5]，[21]，在不同对手角色之间进行泛化的能力[29]，以及在不同游戏之间进行泛化的能力[30]。

这些特质是对文献中各项工作的不同研究目标进行归类后得到的结果。这些都是人工智能体所应具备的理想特质，但并非全部都是严格必需的。我们还认为，这些特质的重要性并不相同。类人性是最值得追求的特征，因为如果缺乏类人性，无论智能体是否展现出其他特质，其带来的游戏体验都会与同人类玩家对战时的体验相去甚远。

以人为中心的智能体应能代表人类玩家。因此，它应覆盖类人策略空间。图1展示了各项特性对人工智能体所覆盖的策略空间的影响。策略空间几何结构基于[31]，中提出的结构，并将在第VI-E节中进一步解释。图中未展示类人性，但它将空间限制在人类玩家可能采用的策略范围内。性能和多样性有助于扩大智能体能够有效覆盖的玩家范围；性能是应对技能水平较高的玩家所必需的，而多样性则是挑战不同游戏风格所必需的。这两项特性分别涉及在传递性与非传递性维度上覆盖的空间。另一方面，难度调整有助于调整智能体的技能水平，而适应性则有助于智能体根据玩家调整自身策略。这两项特性分别涉及在传递性与非传递性维度上遍历空间。最后，从技术角度来看，泛化能力是一项理想特性，因为通用智能体可以在不同角色对局中复用，从而避免在处理所有可能的对局时出现组合爆炸[21]，，并且能够轻松应用于包含不同机制的游戏。它被表示为另一个策略空间，可以对应不同的角色对局，也可以对应完全不同的游戏。这些都是为人类玩家构建智能体时可以考虑的独立特性，不过要全部满足可能较为困难。

开发人工智能体有多种方法，其中最基础的是人工智能（AI）方法，即手动创建规则来决定智能体的行为。然而，这些方法需要大量的专家知识和人力投入。另一种方法是使用机器学习（ML）方法，通过自主学习行为或规则来完成给定任务。还可以使用进化算法与规划算法，这些算法可能更容易适配格斗游戏领域，因为它们通常较少依赖对当前游戏状态的评估，因此对具体问题的依赖也更少。它们可以被整合到其他算法中，这种做法通常是为了提升性能。例如，我们可以将长期规划与反应性更强的算法相结合，如强化学习（RL）[32]，[33]，，也可以将其与进化算法相结合[34]，[35]，，或者仅将进化算法与一般的人工神经网络配合使用[22]，[36]，[37]，[38]。

我们对格斗游戏中人工对手开发的现有技术进行了综述，详细阐述了这些技术在实现以人为中心的智能体各项特质方面取得的成果。本文其余部分的结构如下。第II节更详细地介绍格斗游戏类型。第III节详细介绍本综述采用的方法。第IV节和第V节分别介绍现有的商业游戏和学习环境，详细阐述它们在格斗游戏人工智能体开发方面取得的进展。第VI-A节至第VI-F节讨论与前文列出的以人为中心的智能体各项特质相关的研究。第VII节总结各方法对实现以人为中心的智能体各项特质的贡献。最后，第VIII节总结全文，并指出未来的研究方向。

II. 背景

格斗游戏是一类竞技电子游戏，玩家在竞技场中相互对战[39]。这类游戏的著名代表包括《街头霸王》系列。玩家拥有一定的生命值，目标是在规定时间内耗尽对手的生命值，或在时间结束时保持生命值优势。格斗游戏中有众多角色，每位玩家控制其中一个角色。每个角色都有一套特定的攻防招式，玩家必须运用这些招式才能获胜。图3展示了一个游戏状态示例

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in DareFightingICE, in which each player is using a different character.

A character's moves are performed through button presses or, in rare cases, button releases. This is the way human players interface with these games, and we can consider these button interactions to compose the most basic form of an "action space", i.e., the set of actions that the player can perform. Artificial agents can, of course, deviate from this action space and simplify it. For conciseness, whenever we mention "action" we refer to the basic action space described earlier, unless stated otherwise, e.g., as "macro actions." The complexity with which moves are performed can range from simply performing an action to having to perform a combination or sequence of actions. Moves requiring sequences of actions are usually called "special moves," and tend to be unique to each character. The actions can be subdivided into those focused on movement and those focused on attacking the opponent.

One other prominent feature of fighting games is the ability to perform "combos." A combo is a sequence of offensive moves performed with a short time interval between moves, such that if it hits the opponent it leaves no gaps for them to act back [40]. Since the release of *Street Fighter II*, combos have been a standard mechanic in fighting games, and are an important ability to learn in order to optimize play. Artificial agents should thus have the ability to perform them, which requires learning to perform sets of particular sequences of moves, unless these sequences are included in the agent's action space as actions.

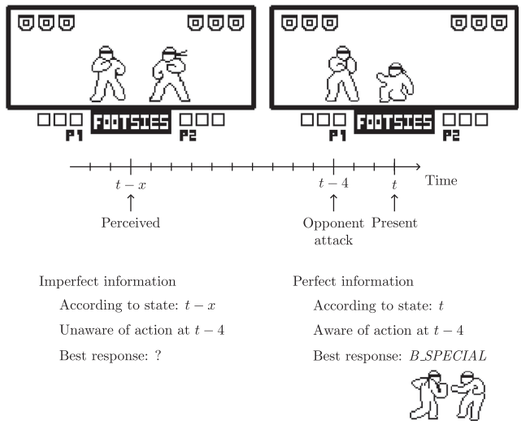


Fig. 2. Influence of perfect information on decision making in fighting games. If players are perfectly informed of all states and actions up to the present, they can perform *B_SPECIAL* in response to any attack, including itself. In that case, players should simply wait for the opponent to attack, since attacking will leave the opponent vulnerable. In this scenario, no player is incentivized to attack, stalling the game. Because human players cannot react to these attacks, this strategy is infeasible.

A. Strategy

Fighting games have imperfect information, unlike many board games. That is, each player does not have full knowledge of the current and previous states when making a decision. Considering chess as an example, each player has access to the full state of the board at all times, which is all that is required to make appropriate decisions. This contrasts with fighting games: these are real-time games with asynchronous moves from both players, and since most of these moves are too fast for a human being to react to, players can never make instantaneous decisions based on those moves. That is, at each instant players have to make decisions without being sure of what move the opponent is doing at that exact moment, and so without knowing the whole game state [34]. As such, it requires predicting what the opponent will do. Because of this, a fair artificial agent should emulate this reaction time and thus rely on prediction as well as reaction, or else it would not be playing by the same game rules as a human player [17], [25], [41], [42].

Fighting games only have perfect information when each player can perceive the current state and act on it instantaneously, which corresponds to a naive implementation of fighting game agents. In this scenario, however, these games lose their strategic appeal. Fig. 2 shows an example of a possible situation in the fighting game *FOOTSIES* [43]. The current state at time step t can be perceived instantaneously by an artificial agent, but human players can only reliably act on state that occurred on a variable number x of time steps earlier. At $t-4$, player two on the right side of the screen began attacking. Attacks have some duration, and at time t player two is still performing the initial portion of the attack, and will only hit the player in the next time step. Informed with this knowledge, player one can perform a particular special move, internally identified as *B_SPECIAL* in the game. This move is invulnerable at the beginning and hits early, allowing player one to counter-attack player two's move. However, its main weakness is that it takes long to finish, which leaves the player vulnerable in case it misses or is defended against. If players can always react instantaneously, then they can perform this move with the guarantee that it will hit, greatly diminishing its weakness, and because of this, the game becomes trivialized. On the other hand, reacting this fast is impossible for human players, and in that case the move carries some risk. This example illustrates how fighting games are fundamentally different from the human perspective: they require players to rely on predictions of the future to compensate for the reaction time delay.

Because of the prediction-based play and nontransitivity of moves, fighting games can be thought of as multiple consecutive trials of *Rock–Paper–Scissors* [6]. Unlike *Rock–Paper–Scissors* however, the trials are not independent of each other, i.e., the outcome of one trial influences the trials performed in the future, since fighting games are prolonged in time. In addition, each move does not yield the same risk or reward, and has long-term consequences that influence the reward that that move is expected to bring in the future. Finally, since each player's moves can be executed at an arbitrary time and take some time to perform, these *Rock–Paper–Scissors* trials in fighting games are extended in time rather than being decided in an instant.

In a competitive game featuring multiple players, a Nash equilibrium is a situation, in which there is no other strategy that could be employed by each player that is better than their current

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在DareFightingICE中，每位玩家使用不同的角色。

角色的招式通过按键按下来执行，少数情况下则通过按键释放来执行。这是人类玩家与这些游戏交互的方式，我们可以将这些按键交互视为构成“动作空间”的最基本形式，即玩家能够执行的动作集合。当然，人工智能体可以不采用这一动作空间，而是对其进行简化。为简洁起见，除非另有说明，例如明确称为“宏动作”，否则文中提到的“动作”均指前述基本动作空间中的动作。执行招式的复杂程度各不相同，既可能只需执行一个动作，也可能需要执行动作组合或动作序列。需要动作序列才能执行的招式通常称为“特殊招式”，而且往往是各个角色独有的。动作可细分为侧重移动的动作和侧重攻击对手的动作。

格斗游戏的另一个突出特点是能够执行“连招”。连招是以较短的招式间隔连续执行的一系列进攻招式，一旦命中对手，就不会给对手留下还击的空隙 [40]。自街头霸王*II*发布以来，连招已成为格斗游戏的标准机制，也是优化游戏表现所需掌握的一项重要能力。因此，人工智能体也应具备执行连招的能力，这要求其学会执行一组特定的招式序列，除非这些序列已作为动作纳入智能体的动作空间。

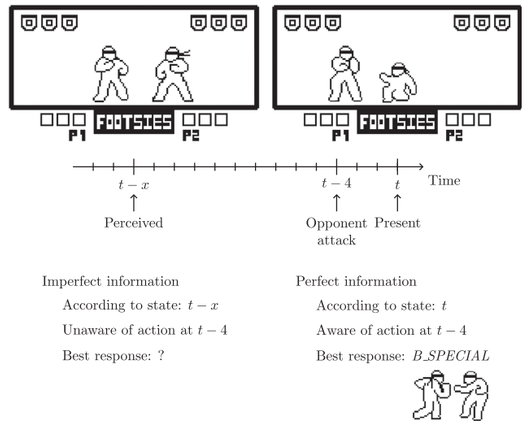


图2. 完全信息对格斗游戏中决策的影响。如果玩家完全掌握截至当前的所有状态和动作信息，就可以施展*B_*特殊招式 来应对任何攻击，包括该招式本身。在这种情况下，玩家只需等待对手发动攻击，因为对手一旦攻击就会暴露破绽。在这一情境下，任何玩家都没有主动攻击的动机，从而导致游戏陷入拖延。由于人类玩家无法对这些攻击及时作出反应，因此这种策略并不可行。

A. 策略

与许多棋盘游戏不同，格斗游戏具有不完美信息。也就是说，每位玩家在做决策时，都无法完全了解当前和先前的状态。以国际象棋为例，每位玩家始终都能获取棋盘的完整状态，而这就是做出恰当决策所需的全部信息。格斗游戏则不同：它们是实时游戏，双方玩家均可施展异步招式，而且由于其中大多数招式的速度快到人类无法及时反应，玩家根本无法根据这些招式即时做出决策。也就是说，玩家在每一瞬间都必须做出决策，却无法确定对手在那一刻正在施展什么招式，因此也就无法了解完整的游戏状态 [34]。因此，玩家需要预测对手接下来会做什么。正因如此，公平的人工智能体应当模拟人类的这种反应时间，从而同时依靠预测和反应，否则它就没有遵循与人类玩家相同的游戏规则 [17], [25], [41], [42]。

只有当每位玩家都能瞬时感知当前状态并据此采取行动时，格斗游戏才具有完全信息，这对应于格斗游戏智能体的一种朴素实现。然而，在这种情况下，这些游戏就失去了策略上的吸引力。图 2 展示了格斗游戏 *FOOTSIES* [43] 中一种可能出现的情境。人工智能体可以瞬时感知时间步 t 的当前状态，但人类玩家只能根据若干时间步之前的状态可靠地采取行动，这一时间步数 x 是可变的。在 $t-4$时，屏幕右侧的玩家二开始攻击。攻击会持续一段时间，而在时刻 t，玩家二仍处于攻击的起始阶段，要到下一个时间步才会击中对方玩家。掌握这一信息后，玩家一可以施展一种特定的特殊招式，其在游戏中的内部标识为 *B_SPECIAL* 。这一招式在起始阶段具有无敌效果，且能较早命中，因此玩家一可以借此反击玩家二的招式。然而，它的主要弱点是收招时间长，一旦未命中或被防御，玩家就会处于易受攻击的状态。如果玩家始终能够瞬时反应，就可以在确保命中的情况下施展这一招式，从而大幅削弱其弱点，游戏也因此变得毫无挑战性。另一方面，人类玩家不可能反应得如此之快，因此对他们而言，这一招式仍具有一定风险。这个例子说明，从人类视角来看，格斗游戏存在根本性的不同：它们要求玩家依靠对未来的预测来弥补反应时间延迟。

由于基于预测的玩法和招式的非传递性，格斗游戏可以视为连续进行的多轮石头剪刀布 [6]。然而，与石头剪刀布不同的是，各轮之间并非相互独立，即一轮的结果会影响后续各轮，因为格斗游戏会在时间上持续展开。此外，每种招式带来的风险或收益各不相同，并且会产生长期影响，进而影响该招式未来的预期收益。最后，由于每位玩家都可以在任意时刻施展招式，且招式的施展需要一定时间，因此格斗游戏中的这些石头剪刀布对局是在一段时间内展开的，而非在瞬间决出结果。

在有多名玩家参与的竞技游戏中，纳什均衡是指这样一种状态：对每位玩家而言，都不存在优于其当前

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one. A Nash equilibrium exists for *Rock–Paper–Scissors*, which involves players randomly choosing any move uniformly [44]. However, fighting games are more complex than *Rock–Paper–Scissors*, and uniform random play may not constitute a Nash equilibrium. An analysis of the Nash equilibrium for fighting games has been done [7], suggesting the Nash equilibrium is indeed a mixed strategy involving stochastic play, but it is incomplete as ample player positioning and the duration of moves were not considered. Still, due to their nontransitivity, we can expect Nash equilibria for fighting games to still be composed of mixed strategies, so that they are not exploitable by being predictable.

However, as is in the *Rock–Paper–Scissors* game, if we are facing an opponent with a specific strategy outside an equilibrium, we can gain an advantage by attempting to use a counter-strategy, yielding greater return than when the equilibrium is followed [45]. In fact, human players tend to deviate from random play in order to exploit patterns in the opponent's behavior [45], [46], [47]. Even when attempting to employ stochastic strategies, human players tend to have biases that drive them to choose actions not randomly but according to some subconscious pattern [8], [31]. This means that, against a specific human player, a Nash equilibrium may not represent the optimal situation and, as such, a method that adapts to the opponent's biases will be more appropriate. Still, the objective of achieving the best performance possible is the one most explored in the literature, detailed further in Section VI-A. The objective of adapting to the current player is explored by works in dynamic difficulty adjustment (DDA), detailed further in Section VI-D.

B. Challenges

This genre poses the following challenges in implementing a human-centered artificial agent.

1) Decision time is standardized to 1/60th of a second, since most of these games process game state at a fixed rate of 60 frames per second.
2) The games are prediction-based rather than reaction-based. Human players cannot react quickly enough to many of the moves that the opponent can make, and these games are built with that in mind. As such, artificial agents should take this into account.
3) Specific sequences and combinations of actions yield specific moves. This increases input complexity, as action sequences require short-term planning, and action combinations increase the effective number of distinct decisions and require multiple action predictors.
4) An ability to adapt to the opponent's behavior quickly is paramount. As explained previously, playing optimally against a human opponent requires exploiting biases in decision making.
5) Combos require long-term planning of moves, which may themselves require planning at the level of actions if the moves need a sequence of actions to be executed.

As such, a human-centered agent for fighting games should exhibit consistent and efficient decision making, be prediction-based, be able to execute sequences and combinations of actions, be quickly adaptive and perform long-term planning. The input complexity can be slightly alleviated by considering combinations and special moves to be actions themselves, although it makes the action space dependent on the character.

III. Methodology

The survey began with building a broad collection of papers related to the fighting game genre, regardless of the work contained therein, to get an overview of research in the area. This brought to our attention some papers that delved into useful concepts, such as [6], [7], even though they are not strictly related to the development of artificial agents. These papers were still important to deepen our understanding of the genre, and proved useful in contextualizing the problem. This search was performed using the Google Scholar, Scopus, and IEEE Xplore search engines. Preference was given to the most recent papers, and results were deemed relevant through analysis of their abstract.

This initial collection of papers was registered in a knowledge base and then reviewed. Whenever we found references of interest while reading the papers, we would read their abstracts and determine if they were relevant for our survey, and if so, include them in the collection. This is akin to the backward snowballing approach for systematic literature reviews [48]. In addition, whenever there was a topic of interest that we found worth researching further, we would broaden our scope and use the aforementioned search engines to find papers related to that topic. Examples of such papers are [31] and [49].

The papers were categorized according to the problem and method that they employed. The list of categories was built incrementally as new papers were reviewed and we felt more categories had to be included or others had to be excluded. The final list of problem categories was "diverse play styles," "human likeness," "adaptability," "DDA," "generalizability," and "opponent modeling." The final list of method categories was "RL," "supervised learning (SL)," "planning-based," "evolutionary algorithm," "rule-based," and "imitation-based." We additionally employed two other categories: "DareFightingICE" includes papers that used the DareFightingICE/FightingICE platform, detailed in Section V-A, and "game theory/psychology" includes all papers that dealt with either game theory, game design or psychology. This categorization proved helpful, as we were able to cluster the papers according to problem-method category pairs.

The bulk of the search was performed in the last quarter of 2023. Since then, the collection of papers was incrementally updated as this survey was being written and our knowledge on the subject improved. Whenever during searches for information we found good candidates for papers to review that we missed or were published after 2023, we included them in the collection. The search was repeated in 2025 to try making sure we did not miss papers from 2024.

IV. Commercial Games

Almost all commercial fighting games either do not employ AI systems exhibiting the previous qualities or are not transparent

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策略的其他可用策略。石头剪刀布存在一种纳什均衡策略，即玩家以均等概率随机选择任一招式 [44]。然而，格斗游戏比石头剪刀布更为复杂，均匀随机对弈可能并不构成纳什均衡。已有研究对格斗游戏的纳什均衡进行了分析 [7]，，表明其纳什均衡确实是一种涉及随机对弈的混合策略，但由于未充分考虑玩家站位和招式持续时间，该分析并不完整。尽管如此，鉴于格斗游戏的非传递性，我们仍可预期其纳什均衡由混合策略构成，从而避免因可预测而被对手利用。

然而，正如在石头剪刀布游戏中，如果我们面对的对手采用了偏离均衡的特定策略，就可以尝试使用反制策略来取得优势，获得比遵循均衡策略时更高的回报 [45]。事实上，人类玩家往往会偏离随机对弈，以利用对手行为中的规律 [45], [46], [47]。即使尝试采用随机策略，人类玩家也往往存在偏置，促使他们并非随机地选择行动，而是遵循某种潜意识中的模式 [8], [31]。这意味着，在面对特定的人类玩家时，纳什均衡可能并不代表最优情形，因此，能够适应对手偏置的方法会更加合适。尽管如此，实现尽可能高的性能仍是文献中研究最多的目标，第VI-A节将对此作进一步介绍。动态难度调整（动态难度调整）领域的研究探索了适应当前玩家这一目标，第VI-D节将对此作进一步介绍。

B. 挑战

在此类游戏中实现以人为中心的人工智能体面临以下挑战。

1) 决策时间统一为 1/60 秒，因为此类游戏大多以每秒 60 帧的固定速率处理游戏状态。2) 这类游戏以预测而非反应为基础。对于对手能够使出的许多招式，人类玩家无法足够迅速地作出反应，而这些游戏在设计时就考虑到了这一点。因此，人工智能体也应将这一点纳入考量。3) 特定的动作序列和连招会产生特定的招式。这增加了输入复杂度，因为动作序列需要短期规划，而动作连招会增加实际可选的不同决策的数量，并需要多个动作预测器。4) 快速适应对手行为的能力至关重要。如前所述，要在与人类对手的对战中采取最优打法，就需要利用其决策中的偏置。5) 连招需要对招式进行长期规划，而如果这些招式需要通过一系列动作来执行，那么招式本身可能也需要动作层面的规划。

因此，用于格斗游戏的以人为中心的智能体应展现出一致且高效的决策能力，以预测为基础，并能够执行动作序列和连招，能够快速适应并进行长期规划。将连招和特殊招式本身视为动作，可以略微降低输入复杂度，但这会使动作空间依赖于角色。

III. 方法

本综述首先广泛收集与格斗游戏类型相关的论文，不论其具体研究内容如何，以了解该领域的研究概况。这使我们注意到了一些深入探讨有用概念的论文，例如 [6], [7], ，尽管这些论文并不严格涉及人工智能体的开发。这些论文对于加深我们对该游戏类型的理解仍然十分重要，并有助于阐明问题的背景。此次检索使用了谷歌学术、Scopus 和 IEEE Xplore 搜索引擎。我们优先考虑最新的论文，并通过分析摘要来判断检索结果的相关性。

我们将初步收集的论文录入知识库，随后进行审阅。在阅读论文时，每当发现感兴趣的参考文献，我们都会阅读其摘要，判断其是否与本综述相关；如果相关，便将其纳入论文集。这类似于系统性文献综述中的后向滚雪球法 [48]。此外，每当发现值得进一步研究的主题时，我们都会扩大检索范围，使用上述搜索引擎查找与该主题相关的论文。此类论文的例子包括 [31] 和 [49]。

我们根据论文所研究的问题和采用的方法对其进行分类。在审阅新论文的过程中，当我们认为需要增加或删除某些类别时，便逐步调整类别列表。最终的问题类别列表为“多样化游戏风格”“类人性”“适应性”“动态难度调整”“泛化能力”和“对手建模”。最终的方法类别列表为“强化学习”“监督学习（SL）”“基于规划”“进化算法”“基于规则”和“基于模仿”。我们还采用了另外两个类别：“DareFightingICE”包括使用 DareFightingICE/FightingICE平台的论文，该平台详见第V-A节；“博弈论/心理学”包括所有涉及博弈论、游戏设计或心理学的论文。这种分类方式很有帮助，使我们能够按照问题与方法的类别组合对论文进行聚类。

大部分检索工作于2023年第四季度完成。此后，随着本综述的撰写以及我们对该主题认识的加深，所收集的论文也在逐步更新。在检索信息的过程中，每当发现此前遗漏或于2023年之后发表、且适合纳入综述的论文时，我们都会将其加入论文集。2025年，我们再次进行了检索，以尽量确保没有遗漏2024年的论文。

IV. 商业游戏

几乎所有商业格斗游戏要么没有采用具备上述特质的人工智能系统，要么未公开

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about their implementation. Of note, the 2013 release of *Killer Instinct* contains a "Shadow System," which creates artificial opponents based on play data of human players [41]. An artificial opponent from this system has a replay bank of matches played by a specific human player, and during inference the agent chooses the moment in the replay bank that most closely resembles the current situation according to a similarity metric. This similarity metric includes heuristics that allow incorporating historical data into the current state, requiring some expert knowledge. After the moment in the bank is chosen, the agent imitates the human player's actions from that moment forward, until a significant state change is detected.

This method is an instance of case-based reasoning, in which an AI system utilizes solutions to past problems to solve new similar problems. Although decision making is reduced to copying past human experience, the method ended up working well in practice with a positive reception from the players [41]. The agents are also easy to train, requiring few samples of human play data to exhibit seemingly meaningful and complex behaviors [41]. Because human play is copied, the agents also inherently retain human limitations, such as reaction time and execution imperfection. The method also has the added benefit of adapting to the players' strategies as they evolve, by simply prioritizing more recent play data, without needing any further tuning.

However, the AI from the Shadow System does not explicitly perform decisions, but simply imitates other players. This means that the AI is not necessarily playing to win. If the human player it is imitating is losing on purpose, then it will lose on purpose as well. The AI does not have any notion of how good or bad each decision is, i.e., the replay moment it chooses, in terms of future reward. This allows the AI to exhibit all kinds of behaviors including noncompetitive ones, such as taunting, but not to autonomously provide insight into how balanced the game is, for instance. It also does not play with the intention of countering the player's strategy directly, which has been found to be a limitation of case-based reasoning in the literature [50]. We want to note that case-based reasoning is not to be confused with "imitation learning," which is a paradigm in RL focused on a similar goal, but encompasses a different set of methods, such as SL and inverse RL [51].

Another example worth detailing is *Street Fighter 6*, released in 2023. The developers designed its AI with human-centered features in mind, mainly human likeness and difficulty adjustment [52]. Their intent was to create an artificial opponent that can aid a player in learning the mechanics of the game, convey the particularities of a character, and help determine a player's skill level before they use skill-based online matchmaking. This is a human-centered approach that aims to improve the player experience by providing agents that play similarly to human players. In addition, the agent needs to be useful for players of different skill levels. To do so, the developers incorporate three main components in the agent: abstractions for the agent's action space, an emulation of the feeling of pressure and variable knowledge of the game. By tuning these three components, we can create agents of different difficulty levels.

Street Fighter 6's agents do not use the basic action space where each action is a button interaction. Instead, sequences of actions are abstracted into macro actions. This is similar to the way human players make decisions; rather than thinking in terms of button presses or releases, human players decide on a move or sequence of moves to perform. Each action in an action space has information associated with it that influences the decision, such as the damage and speed of the move. The action space is different for each character, which is necessary since the moves of each character may require different sequences of actions to perform, and characters encourage different play styles. In addition, the complexity of the action space varies with the desired difficulty of the agent, with harder agents including more features of the game such as special moves. Finally, the precision of actions changes with difficulty: on easier difficulties, the agent is prone to making mistakes when executing a macro action, i.e., pressing one button instead of another, akin to the mistakes done by human players. Decision making over this action space is performed using finite-state machines. Care was taken to make the AI dynamic, i.e., rely as much as possible on the current state rather than predefined information. This allows the AI to be resilient to changes in the game's balance, that occur over time.

The AI in *Street Fighter 6* contains artificial handicaps to emulate characteristics of human players. A pressure coefficient is calculated during the match that influences the precision with which an agent performs actions. The lower the difficulty, the more the agent is affected by pressure. Another factor that is incorporated is imperfect knowledge. The agent has three different levels of knowledge about whether it is advantageous in a particular state, i.e., whether it can act before the opponent can: does not know, knows whether it is advantageous, and knows exactly by *how much* it is advantageous, in terms of frames. For lower levels of difficulty, the agent has less knowledge about advantage, which affects decision making. Tuning these components allows modulating the difficulty of the agent in a manner that is consistent with how skill varies in human players, making it able to exhibit human-like behavior accross difficulty levels.

There are other noteworthy examples that, although revealing their general approach, were not fully transparent on the solution. *Samurai Shodown* [53], released in 2019, uses deep learning (DL) for its AI. This system, similar to *Killer Instinct*'s, imitates human players and creates "Ghost" characters for each one. This allows human players to emulate playing against themselves, as well as play against artificial agents trained on other players. However, details on the system's architecture or reception by the players have not been disclosed. *Tekken 8*, released in 2024, features a "Super Ghost Battle" mode that allows each player to build an AI that learns to imitate them, akin to the above examples. The AI utilizes RL to learn, more specifically the Q-learning algorithm [54]. However, no more details have been disclosed, and it is not clear whether the reward for the Q-learning algorithm is a reward for closely imitating the human player or for playing the game successfully, only utilizing the human play data to know what experience to use for the updates. Also in 2024, *Street Fighter 6* introduced the "V-Rival" system [55], which is similar in goal to *Killer Instinct*'s Shadow

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其实现方式。值得注意的是，2013年发布的《杀手本能》包含一个"暗影系统"，该系统根据人类玩家的游戏数据创建人工对手 [41]。该系统中的人工智能对手拥有一个回放库，其中收录了某位特定人类玩家的对局；在推理过程中，智能体会依据相似性度量，从回放库中选择与当前情境最为相似的时刻。这种相似性度量包含一些启发式规则，可将历史数据纳入当前状态，因此需要一定的专家知识。选定回放库中的时刻后，智能体便从该时刻起模仿人类玩家的操作，直到检测到显著的状态变化。

该方法是基于案例的推理的一个实例，即人工智能系统利用过去问题的解决方案来解决新的类似问题。尽管决策被简化为复制过去的人类经验，但该方法在实践中最终取得了良好效果，并获得了玩家的积极评价 [41]。这些智能体也易于训练，仅需少量人类游戏数据样本，就能展现出看似有意义且复杂的行为 [41]。由于复制了人类的游戏行为，这些智能体也自然保留了人类的局限性，例如反应时间和执行不完美性。该方法还有一个额外优势：只需优先使用较新的游戏数据，就能适应玩家不断演变的策略，而无需进行任何进一步调优。

然而，暗影系统中的人工智能并不会显式地进行决策，而只是模仿其他玩家。这意味着，该人工智能并不一定以获胜为目标。如果它所模仿的人类玩家故意输掉比赛，那么它也会故意输掉比赛。该人工智能并不了解每项决策（即它所选择的回放片段）在未来奖励方面的好坏。这使得人工智能能够展现各种行为，包括嘲讽等非竞技性行为，但却无法自主地提供见解，例如评估游戏的平衡性。它在游戏中也不会以直接克制玩家策略为目标，而文献已指出，这是基于案例的推理的一项局限 [50]。我们想指出，基于案例的推理不应与"模仿学习"混淆，后者是强化学习中的一种范式，关注类似的目标，但涵盖的是一组不同的方法，例如监督学习和逆强化学习 [51]。

另一个值得详细介绍的例子是街头霸王*6*，该游戏于2023年发行。开发者在设计其人工智能时，着重考虑了以人为中心的特性，主要包括类人性和难度调整 [52]。他们旨在打造一种人工智能对手，帮助玩家学习游戏机制、了解角色的特点，并在玩家使用基于技能的在线匹配功能之前，协助确定其技能水平。这是一种以人为中心的方法，旨在通过提供游戏行为与人类玩家相似的智能体来改善玩家体验。此外，智能体还需要对不同技能水平的玩家有所帮助。为此，开发者在智能体中引入了三个主要组成部分：智能体动作空间的抽象、对压力感的仿真，以及对游戏知识掌握程度的调整。通过调节这三个组成部分，可以创建不同难度等级的智能体。

街头霸王*6*中的智能体并不使用每个动作都对应一次按键交互的基础动作空间，而是将动作序列抽象为宏动作。这与人类玩家的决策方式相似：人类玩家并不会从按键按下或按键释放的角度思考，而是决定要执行哪种招式或招式序列。动作空间中的每个动作都附带影响决策的信息，例如招式的伤害和速度。每个角色的动作空间各不相同，这是必要的，因为各个角色的招式可能需要通过不同的动作序列来执行，而且不同角色也会引导玩家采用不同的游戏风格。此外，动作空间的复杂度会随智能体的目标难度而变化，难度更高的智能体会纳入更多游戏要素，例如特殊招式。最后，动作精度也会随难度而变化：在较低难度下，智能体在执行宏动作时容易出错，即误按其他按键，这与人类玩家所犯的错误类似。针对这一动作空间的决策通过有限状态机实现。开发者特别注重使人工智能具有动态性，即尽可能依赖当前状态，而非预定义信息。这使人工智能能够适应游戏平衡性随时间推移而发生的变化。

街头霸王*6* 中的人工智能设置了人为限制，以模拟人类玩家的特征。比赛过程中会计算一个压力系数，该系数会影响智能体执行动作的精确度。难度越低，智能体受压力的影响就越大。另一个纳入考虑的因素是不完全知识。对于自身在特定状态下是否具有优势，即能否先于对手行动，智能体具有三个不同的认知层次：不知道、知道是否具有优势，以及确切知道优势有多大（以帧为单位）。在较低的难度等级下，智能体对优势的了解较少，这会影响其决策。调整这些组成要素，可以按照与人类玩家技能差异相一致的方式调节智能体的难度，使其能够在不同难度等级下展现类人行为。

还有一些值得关注的例子，虽然披露了总体方法，但并未完全公开具体解决方案。侍魂 [53], 于2019年发行，其人工智能采用深度学习（DL）。该系统与杀手本能的系统类似，会模仿人类玩家，并为每位玩家创建"幽灵"角色。这使人类玩家既能模拟与自己对战，也能与基于其他玩家训练的人工智能体对战。然而，其系统架构的细节及玩家的反响均未披露。铁拳*8*于2024年发行，提供了"超级幽灵对战"模式，与上述例子类似，每位玩家都可以构建一个通过学习来模仿自己的人工智能。该人工智能利用强化学习进行学习，更具体地说，采用的是Q学习算法 [54]。然而，尚无更多细节披露，也不清楚Q学习算法中的收益究竟是针对高度模仿人类玩家的行为，还是针对成功进行游戏的表现，而人类游戏数据仅用于确定更新时应使用哪些经验。同样在2024年，街头霸王*6*推出了"V-Rival"系统 [55],，其目标与杀手本能的暗影

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TABLE I
COMPARISON BETWEEN AI SYSTEMS IN COMMERCIAL FIGHTING GAMES

System	Game	Year	Goals	Method	Implementation
Shadow System	*Killer Instinct*	2013	Capture the experience of playing with a human player Be resilient to game updates Create, share, and play with imitations of human players	Mimicry	Case-based reasoning
Figure Players	*Super Smash Bros. 4* *Super Smash Bros. Ultimate*	2014 2018	Create, share, and play with imitations of human players	—	—
Ghost	*Samurai Shodown*	2019	Create, share, and play with imitations of human players	—	Deep learning
In-game AI	*Street Fighter 6*	2023	Capture the experience of playing with a human player Be resilient to game updates Be tunable in order to produce different difficulty levels	Rule-based	Finite-state machines
Super Ghost Battle	*Tekken 8*	2024	Create, share, and play with imitations of human players	RL	Q-learning
V-Rival	*Street Fighter 6*	2024	Create, share, and play with imitations of human players	—	—

System. Individual player data is collected and used to train artificial agents of that player, that can then be played against by anyone. The matches with V-Rival agents are also analyzed, after which comments and recommendations for improvement are given. However, no details were given on its implementation. The *Super Smash Bros.* series of fighting games feature "Figure Players," which are trainable AI that learn to play similarly to the human player they fight against, scanned and saved through near-field communication on amiibo toys [56]. Like the previous examples, it is not known exactly how these "Figure Players" learn.

Table I compares the aforementioned AI systems in terms of goals, method and implementation. Almost all systems aim to create agents from imitation of human players. Direct imitation of human players allows creating human-like AI, which helps bridge the gap between the experience of fighting in-game AI and other human players, useful to improve learnability. The in-game AI of *Street Fighter 6* stands out as the only one that is not based on imitation, and instead the decision making is entirely engineered by the developers. It also differs from the other examples in terms of providing different difficulty levels; the imitation-based AI can supply agents of varying skill levels if there is play data of players of different skills, whereas *Street Fighter 6*'s AI has tunable parameters. Along with the Shadow System, it was designed in such a way that game balance updates do not require extensive changes on the agents. For *Killer Instinct*'s Shadow System, the AI simply needs to be supplied new play data, while in *Street Fighter 6* the AI relies on data from the current state. For *Samurai Shodown*, while the general approach is known, it is not known which method is used, i.e., SL, RL, etc. To the best of the authors' knowledge, there are no commercial games that employ AI that performs online learning on its own, other than learning to imitate the opponent. Overall, we can note that commercial games tend to apply methods based on imitation for building human-like AI, with *Killer Instinct* and *Street Fighter 6* being those whose method is known in greater detail.

V. LEARNING ENVIRONMENTS

There are not many fighting game environments built specifically for the development of AI, especially since fighting games are mostly commercial games. Still, we identify three platforms offering such environments: DareFightingICE/FightingICE, DIAMBRA Arena, and FightLadder. DareFightingICE was formerly known as FightingICE prior to 2022. Hereafter, we will refer to the platform as a whole as "DareFightingICE" for conciseness, and refer to it as "FightingICE" when we specifically want to mention versions of it prior to 2022.

Table II presents a summary comparison between the identified platforms. DIAMBRA Arena and FightLadder have a large number of environments available, as they emulate existing commercial games. DIAMBRA Arena is the platform that offers the greatest variety in fighting game subgenres, with the other two only providing 2-D nontag fighting game environments. 2-D games allow movement in two dimensions, whereas 3-D games allow players to move into the foreground and background. In tag-based games, players are able to use multiple characters and switch between them on demand during the match, whereas usually players only control one at a time. While DareFightingICE endorses any kind of method, the other two platforms are mainly focused on RL. FightLadder is the environment that officially provides the largest number of pretrained sample AI, although DareFightingICE has a large catalogue of open-source agents used in its competitions. Although DIAMBRA Arena has hosted one competition and has a live leaderboard, the submitted agents are not available to the public. DareFightingICE is the only platform providing audio features, and although at one point also supported the Gym API [57], it was deprecated. In terms of modifications on the basic action space described in Section II, DareFightingICE and FightLadder discretize special moves, i.e., include the required sequence and combinations of actions as a macro action. DIAMBRA Arena and FightLadder also provide multi discrete action spaces, where movement or defensive actions are separated from offensive actions and can be performed simultaneously.

The following sections describe each platform in more detail.

A. DareFightingIce

DareFightingICE is a fighting game made for research of AI algorithms using any methodology, released in 2013. This platform was developed by the Intelligent Computer Entertainment Lab[1] at Ritsumeikan University, and is illustrated in Fig. 3.

[1]https://www.ice.ci.ritsumei.ac.jp/links.html

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表I 商业格斗游戏中的人工智能系统比较

System	Game	Year	Goals	Method	Implementation
Shadow System	*Killer Instinct*	2013	Capture the experience of playing with a human player Be resilient to game updates Create, share, and play with imitations of human players	Mimicry	Case-based reasoning
Figure Players	*Super Smash Bros. 4* *Super Smash Bros. Ultimate*	2014 2018	Create, share, and play with imitations of human players	—	—
Ghost	*Samurai Shodown*	2019	Create, share, and play with imitations of human players	—	Deep learning
In-game AI	*Street Fighter 6*	2023	Capture the experience of playing with a human player Be resilient to game updates Be tunable in order to produce different difficulty levels	Rule-based	Finite-state machines
Super Ghost Battle	*Tekken 8*	2024	Create, share, and play with imitations of human players	RL	Q-learning
V-Rival	*Street Fighter 6*	2024	Create, share, and play with imitations of human players	—	—

系统相似。系统会收集各个玩家的玩家数据，并用于训练代表该玩家的人工智能体，随后任何人都可以与其对战。系统还会分析与V-Rival智能体的对局，并据此给出评价和改进建议。然而，其具体实现细节并未公开。任天堂明星大乱斗系列格斗游戏设有“模型斗士”，这是一种可训练人工智能，能够学习与其对战的人类玩家的玩法，并通过近场通信在amiibo玩偶上进行扫描和保存 [56]。与前述例子一样，这些“模型斗士”究竟如何学习，目前仍不清楚。

表I 从目标、方法和实现方面比较了上述人工智能系统。几乎所有系统都旨在通过模仿人类玩家来创建智能体。直接模仿人类玩家可以创建类人人工智能，有助于缩小与游戏内人工智能对战和与其他人类玩家对战之间的体验差距，从而提高易学性。街头霸王*6*的游戏内人工智能是其中唯一不基于模仿的系统，其决策完全由开发者设计。它在提供不同难度等级方面也与其他示例有所不同：基于模仿的人工智能在拥有不同技能玩家的游戏数据时，可以提供不同技能水平的智能体，而街头霸王*6*的人工智能则具有可调参数。与暗影系统一样，它在设计时便考虑到游戏平衡性更新不应要求对智能体进行大幅修改。对于杀手本能的暗影系统，只需向人工智能提供新的游戏数据；而在街头霸王*6*中，人工智能依赖当前状态的数据。对于侍魂，虽然其总体思路已知，但具体采用何种方法（如监督学习、强化学习等）尚不明确。据作者所知，除学习模仿对手外，尚无商业游戏采用能够自主进行在线学习的人工智能。总体而言，我们可以看到，商业游戏倾向于采用基于模仿的方法来构建类人人工智能，其中杀手本能和街头霸王*6*所采用的方法已有较为详细的公开信息。

V. 学习环境

专为人工智能开发而构建的格斗游戏环境并不多，尤其是因为格斗游戏大多属于商业游戏。尽管如此，我们仍找到了三个平台提供此类环境：DareFightingICE/FightingICE、DIAMBRAArena 和 FightLadder。DareFightingICE 在2022年之前名为 FightingICE。下文为简洁起见，将该平台统称为“DareFightingICE”，而在专门提及其2022年之前的版本时，则称其为“FightingICE”。

表 II 汇总比较了已确定的各个平台。DIAMBRA Arena 和 FightLadder 通过模拟现有商业游戏，提供了大量可用环境。DIAMBRA Arena 提供的格斗游戏子类型最为丰富，而另外两个平台仅提供二维非组队格斗游戏环境。二维游戏允许在两个维度上移动，而三维游戏允许玩家向前景和背景移动。在组队换人格斗游戏中，玩家可以使用多个角色，并在比赛期间按需切换，而通常情况下，玩家一次只能控制一个角色。DareFightingICE 支持使用任何类型的方法，而另外两个平台主要侧重于强化学习。FightLadder 是官方提供预训练示例人工智能数量最多的环境，不过 DareFightingICE 也拥有大量用于其竞赛的开源智能体。尽管 DIAMBRA Arena 已举办过一次竞赛，并设有实时排行榜，但提交的智能体并未向公众开放。DareFightingICE 是唯一提供音频特征的平台，虽然它曾一度支持 Gym API， [57], 但该支持已被弃用。在对第 II 节所述基本动作空间的修改方面，DareFightingICE 和 FightLadder 将特殊招式离散化，即把所需的动作序列与连招纳入一个宏动作中。DIAMBRA Arena 和 FightLadder 还提供多离散动作空间，将移动或防御动作与进攻动作分离，使它们能够同时执行。

以下各节将更详细地介绍各个平台。

A. DareFightingIce

DareFightingICE 是一款于2013年发布的格斗游戏，旨在支持采用各种方法的人工智能算法研究。该平台由立命馆大学的智能计算机娱乐实验室开发[1]，如图 3 所示。

[1]https://www.ice.ci.ritsumei.ac.jp/links.html

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TABLE II
COMPARISON BETWEEN LEARNING ENVIRONMENT PLATFORMS FOR THE DEVELOPMENT OF ARTIFICIAL AGENTS FOR FIGHTING GAMES

Feature	DareFightingICE	DIAMBRA Arena	FightLadder
Release year	2013	2021	2024
Number of environments	1	9	5
Environment source	Custom	Commercial	Commercial
Sub genres covered	2-D non-tag	2-D tag/non-tag and 3-D tag/non-tag	2-D non-tag
Main methods	Any	RL, imitation learning	RL
Pre-trained sample agents	3	3, across different games	5 for each screen side, for one game only
Competitions	12	1, and a continuous competition through live leaderboards	—
Participant source code	Yes, up to 2023	No	—
State representation	Structured, visual and audio	Structured and visual	Structured and visual
Action space modifications	Discretized special moves	Multi discrete	Multi discrete, discretized special moves
Interface type	Java library and Python library (gRPC)	Gymnasium API	Gym API

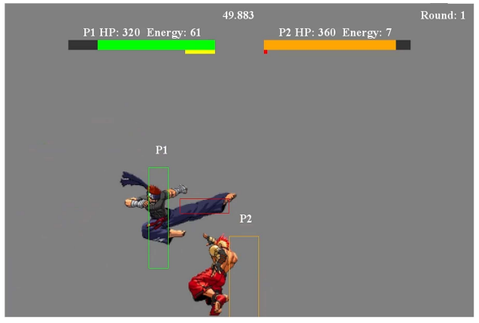


Fig. 3. Screenshot of a round in DareFightingICE, version 6.3.1, between the characters ZEN and LUD.

This game works like usual fighting games, with a framerate of 60 frames per second and specific action sequences activating special moves [58]. Note that these action sequences are only mandatory for human players, and artificial agents can simply execute special moves by activating a sequence of actions. Agents interface with the game through a Java library or, alternatively, a Python library, which communicates with the game through gRPC [59], although the game recently transitioned to using sockets rather than gRPC.

The Intelligent Computer Entertainment Lab hosts yearly competitions since 2013, in which participants develop AI agents for the platform to fight each other in a tournament [60]. The main goal of the competition is to advance research in the development of general AI, specifically for fighting games. A general fighting game AI should be able to adapt to and surpass unseen opponents, using any kind of character that the game offers. Since 2022, the platform changed its name from "FightingICE" to "DareFightingICE," and the competition changed focus to the development and evaluation of sound designs appropriate for gameplay by visually impaired players. The DareFightingICE competitions have two separate sections, one for sound design development and another for agent development using sound as sensory input, which became independent competitions in 2024.

Artificial agents can perceive the game state using three distinct representations: audio, visual, or structured. Audio-based features are a recent addition that has been experimented with in the DareFightingICE competition. It has been proven that, with appropriate sound design, fighting games can be played based on sound cues alone.[2] Utilizing audio-based features for creating artificial agents is also an interesting problem that has not been as extensively explored as using visual-based ones. The game attempts to emulate the reaction time of human players by delaying the structured state information sent to the artificial agents by 15 frames, which corresponds to 250 ms. However, audio and visual data are not delayed.

Three playable characters are offered. The offering of three playable characters, each behaving differently, affords experimenting with generalization.

One important feature of the game is the ability for the agent to perform simulations of the game on demand. This heavily simplifies the application of algorithms dependent on knowing the environment dynamics, since they are already provided. This makes the implementation of planning algorithms, such as Monte Carlo tree search (MCTS) more straightforward, relieving the burden of learning an environment model from the agent. However, simulation is computationally expensive, so the time between time steps (which, as a reminder, is 1/60th of a second) is too short to allow extensive simulation [61], [62]. The ability to learn environment models could allow for the use of more abstract representations of the dynamics that encode information that is not only useful for the agent but is also faster to compute. Consider, for instance, temporally abstracted simulations that allow several frames to be performed at once. Still, learning such a model is an open research question [63].

Fig. 4 shows the evolution of methods used among the competitors for the FightingICE and DareFightingICE competitions over the years. We can broadly classify approaches as being in one of three main categories: rule-based, planning-based, and RL-based [64]. Note that each competitor may combine multiple

[2]Examples at: https://www.ice.ci.ritsumei.ac.jp/~ftgaic/Downloadfiles/2022.pdf

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表 II
比较 用于开发格斗游戏人工智能体的学习环境平台之间 ES

Feature	DareFightingICE	DIAMBRA Arena	FightLadder
Release year	2013	2021	2024
Number of environments	1	9	5
Environment source	Custom	Commercial	Commercial
Sub genres covered	2-D non-tag	2-D tag/non-tag and 3-D tag/non-tag	2-D non-tag
Main methods	Any	RL, imitation learning	RL
Pre-trained sample agents	3	3, across different games	5 for each screen side, for one game only
Competitions	12	1, and a continuous competition through live leaderboards	—
Participant source code	Yes, up to 2023	No	—
State representation	Structured, visual and audio	Structured and visual	Structured and visual
Action space modifications	Discretized special moves	Multi discrete	Multi discrete, discretized special moves
Interface type	Java library and Python library (gRPC)	Gymnasium API	Gym API

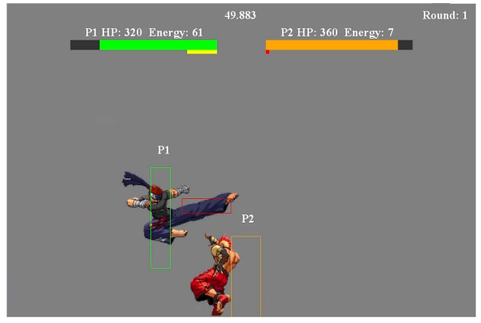


图 3. DareFightingICE 版本6.3.1 中角色 ZEN 与 LUD 之间一局对战的截图。

该游戏的运行方式与常规格斗游戏相同，帧率为每秒60帧，通过特定动作序列触发特殊招式 [58]。需要注意的是，只有人类玩家必须执行这些动作序列，而人工智能体只需激活一系列动作即可执行特殊招式。智能体通过 Java库或 Python库与游戏交互，后者通过 gRPC 与游戏通信 [59]，，不过该游戏最近已改用套接字，不再使用 gRPC。

智能计算机娱乐实验室自2013年起每年举办竞赛，参赛者为该平台开发人工智能体，并让它们在锦标赛中相互对战 [60]。竞赛的主要目标是推动通用人工智能的开发研究，尤其是面向格斗游戏的研究。通用的格斗游戏人工智能应能够使用游戏提供的任意角色，适应并战胜此前未遇到过的对手。自2022年起，该平台由“FightingICE”更名为“DareFightingICE”，竞赛的重点也转向开发和评估适合视障玩家进行游戏的声音设计。DareFightingICE竞赛设有两个独立组别，一个面向声音设计开发，另一个面向以声音作为感官输入的智能体开发；这两个组别于2024年成为独立竞赛。

人工智能体可以通过三种不同的表示形式感知游戏状态：音频、视觉或结构化表示。基于音频的特征是近期新增的内容，已在DareFightingICE竞赛中进行试验。已有研究证明，只要声音设计得当，仅凭声音线索即可游玩格斗游戏。[2] 利用基于音频的特征创建人工智能体也是一个有趣的问题，但相较于利用基于视觉的特征，这方面的探索尚不充分。该游戏通过将发送给人工智能体的结构化状态信息延迟15帧（相当于250毫秒），尝试模拟人类玩家的反应时间。不过，音频和视觉数据不会延迟。

提供三个可玩角色。这三个可玩角色各自具有不同的行为，可用于开展泛化实验。

该游戏的一项重要特性是，智能体能够按需对游戏进行仿真。这大大简化了依赖已知环境动力学的算法的应用，因为游戏已经提供了环境动力学。这使得蒙特卡洛树搜索（MCTS）等规划算法的实现更加直接，免去了智能体学习环境模型的负担。然而，仿真的计算成本很高，因此相邻时间步之间的间隔（提醒一下，该间隔为1/60秒）太短，无法支持大量仿真 [61], [62]。学习环境模型的能力或可支持使用更抽象的动力学表示，这些表示所编码的信息不仅对智能体有用，而且计算起来也更快。例如，时间抽象仿真能够一次推进多个帧。不过，如何学习这样的模型仍是一个尚待解决的研究问题 [63]。

图4 展示了多年来FightingICE竞赛和DareFightingICE竞赛参赛者所采用方法的演变。我们可以将这些方法大致分为三大类：基于规则、基于规划和基于强化学习 [64]。请注意，每位参赛者都可能结合多种

[2]示例见：https://www.ice.ci.ritsumei.ac.jp/~ftgaic/Downloadfiles/2022.pdf

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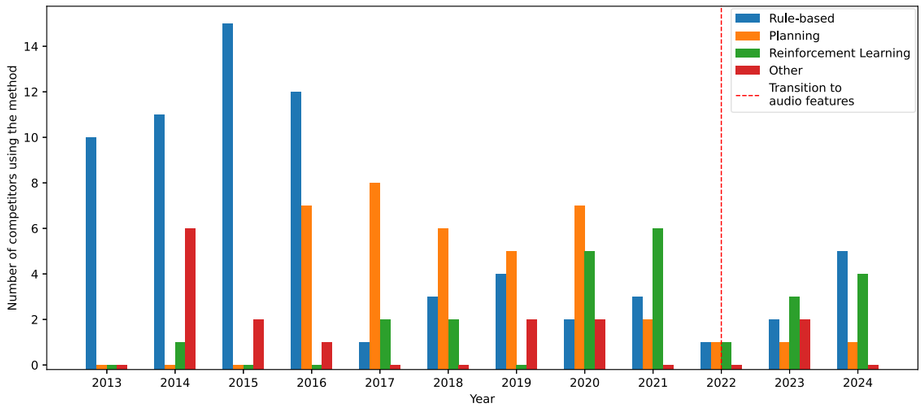


Fig. 4. Method adoption of FightingICE and DareFightingICE competitors over the years. The "other" category includes all works that neither used rules, planning or RL, such as evolutionary algorithms. From 2022, the competition transitioned from the "FightingICE" competition to the "DareFightingICE" competition. The data considers the participants' source code and conference presentations, and includes agents developed by the organization, such as MctsAi.

methods, e.g., planning may be combined with RL, which is common. Methods that are classified into "other" are those that neither used RL, planning nor manual rules such as evolutionary algorithms.

In earlier versions of the competition, mainly rule- and heuristic-based agents were developed. Since 2016, planning using MCTS has been widely adopted among competitors, matching the year AlphaGo achieved a milestone in RL by combining RL with neural networks and MCTS-based planning [65]. MCTS is also particularly useful in fighting games since it can be executed for an arbitrary amount of time and produce partial results. Therefore, the whole limited time budget for every decision can be allotted to planning. However, it may not be a good method for a training opponent for the same reason, as it will make training occur in real time, and speeding it up sacrifices the performance of the training opponent [22].

In recent years, there has been more interest in using SL and RL for agent development. Still, most submissions to the latest competition, in 2024, were rule-based and had good performance, with four such submissions making up the top five. The preference for rule-based agents and their success may be attributed to the transition to audio features, which can still be hard to learn from using SL and RL, as they are not as extensively explored as visual and structured state features. Rule-based agents can also behave more predictably, whereas RL methods are more prone to behaving weirdly in exceptional situations [66]. Adoption of planning has seen a decline. Since 2022, the only planning-based agent was MctsAi with a very low time budget, provided by the competition organizers and the only agent not using audio-based features [60]. All submissions using SL and RL used DL methods, which can tackle large amounts of information. On the other hand, planning methods can be state-agnostic. For example, the most basic implementation of MCTS does not need to analyze the features that make up the current state; all that is required is that actions are simulated on that state. A state can be identified not by its features but by the sequence of actions performed to reach it. As such, a possible reason for the decline is that the platform is focused on the use of audio features, and usage of methods that are agnostic to those features is not endorsed.

Various works use DareFightingICE/FightingICE as a testbed for experimentation. The platform provides a baseline MctsAi that purely plans with MCTS, made available in 2016, and two other baselines based on RL: one using a Deep Q-Network (DQN), introduced in 2018, and another using proximal policy optimization, introduced in 2022. These AIs, mainly the one based on pure MCTS, are frequently used to comparatively evaluate performance both in the competitions and the literature.

B. DIAMBRA Arena

DIAMBRA is an ecosystem comprising two platforms for development of and competition among AI, respectively, focused on RL. Although RL is the focus, the platform also provides functionality supporting imitation learning. DIAMBRA Arena is the development platform, allowing development of agents in competitive video games for research, competition and experimentation [67]. The project has been hosted on GitHub since 2021, and started supporting tournaments on the same year. It provides environments by emulating various commercial games, with the current selection featuring fighting games only. This selection is presented in Fig. 5. The supported games cover different subgenres of fighting games, providing a great variety in game mechanics. Agents interact with the environments through the Gymnasium API [68]. The platform already offers various wrappers and utilities for this API to support different use cases, such as imitation learning.

The DIAMBRA competition platform hosts a continuous competition in which competitors can asynchronously submit agents to the platform as a solution for one of the environments. The platform calculates a score for each submission, and a live leaderboard presents all submissions made to each environment

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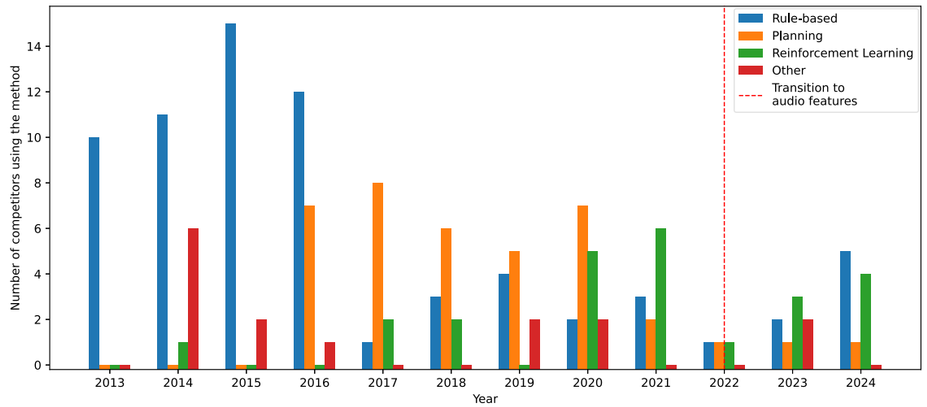


图4. FightingICE和DareFightingICE历年参赛者的方法采用情况。"其他"类别包括所有未使用规则、规划或强化学习的工作，例如采用进化算法的工作。自2022年起，赛事由"FightingICE"竞赛转为"DareFightingICE"竞赛。数据依据参赛者的源代码和会议报告整理，并包含主办方开发的智能体，如MctsAi。

方法，例如，将规划与强化学习相结合，这种做法很常见。归为"其他"的方法是指既不使用强化学习、规划，也不使用人工规则的方法，例如进化算法。

在早期的竞赛中，开发的主要是基于规则和启发式的智能体。自2016年起，使用蒙特卡洛树搜索进行规划的方法在参赛者中得到广泛采用；同年，阿尔法围棋通过将强化学习与神经网络及基于蒙特卡洛树搜索的规划相结合，实现了强化学习领域的一项里程碑 [65]。蒙特卡洛树搜索在格斗游戏中也尤为实用，因为它可以运行任意时长并产生部分结果。因此，每次决策有限的时间预算都可以全部分配给规划。然而，出于同样的原因，这种方法可能并不适合用于训练对手，因为它会使训练只能实时进行，而加快训练速度则会牺牲训练对手的性能 [22]。

近年来，使用监督学习和强化学习进行智能体开发受到越来越多的关注。然而，在2024年最近一届竞赛中，大多数参赛作品仍基于规则，且性能良好，前五名中有四个属于此类作品。基于规则的智能体之所以受到青睐并取得成功，可能是因为竞赛转向了使用音频特征；与视觉与结构化状态特征相比，音频特征尚未得到同等广泛的探索，因此使用监督学习和强化学习从中学习仍可能较为困难。基于规则的智能体的行为也更容易预测，而强化学习方法在异常情况下更容易表现出怪异行为 [66]。规划的采用有所减少。自2022年以来，唯一基于规划的智能体是由竞赛组织者提供、时间预算极低的MctsAi，它也是唯一不使用基于音频的特征的智能体 [60]。所有采用监督学习和强化学习的参赛作品都使用了能够处理大量信息的深度学习方法。另一方面，规划方法可以不依赖于状态。例如，最基本的蒙特卡洛树搜索实现不需要分析构成当前状态的特征；只需在该状态下模拟动作即可。状态可以不通过其特征来识别，而是通过为达到该状态而执行的动作序列来识别。因此，规划采用减少的一个可能原因是，该平台侧重于使用音频特征，并不鼓励使用不依赖这些特征的方法。

多项研究使用 DareFightingICE/FightingICE 作为实验测试平台。该平台提供了一个于2016年推出、完全使用蒙特卡洛树搜索进行规划的基线 MctsAi，以及另外两个基于强化学习的基线：一个使用深度Q网络（DQN），于2018年推出；另一个使用近端策略优化，于2022年推出。这些人工智能，尤其是完全基于蒙特卡洛树搜索的人工智能，经常被用于竞赛和文献中的性能比较评估。

B. DIAMBRA Arena

DIAMBRA 是一个由两个平台组成的生态系统，分别用于人工智能开发和竞赛，重点关注强化学习。尽管以强化学习为重点，该平台也提供支持模仿学习的功能。DIAMBRA Arena 是其中的开发平台，支持开发用于竞技电子游戏的智能体，以开展研究、竞赛和实验 [67]。该项目自2021年起托管于 GitHub，并于同年开始支持锦标赛。它通过模拟各种商业游戏来提供环境，目前所选游戏均为格斗游戏。这些游戏如图5所示。支持的游戏涵盖了格斗游戏的不同子类型，提供了丰富多样的游戏机制。智能体通过 Gymnasium API 与环境交互 [68]。该平台已为此 API 提供多种封装器和实用工具，以支持模仿学习等不同用例。

DIAMBRA竞赛平台举办了一项持续进行的竞赛，参赛者可以异步向平台提交智能体，作为某个环境的解决方案。平台会为每次提交计算得分，实时排行榜则展示各个环境中迄今为止的所有提交

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Fig. 5. Selection of commercial games available as environments in the DIAMBRA Arena platform. The current selection is diverse, covering different fighting game subgenres. Note that the versions of *Marvel versus Capcom: Clash of Super Heroes* and *X-Men versus Street Fighter* available on the platform do support tag-based fighting, contrary to the version of those games released for the *Playstation* console.

thus far, ranked in terms of score. The score is calculated by running the submitted agent for a single episode, and is a function of the cumulative reward and the game difficulty. The difficulty is specified when the agent is submitted, and can be either "easy," "medium," or "hard," with each subsequent level increasing the score obtained for the same amount of cumulative reward. The score-amplifying effect of each difficulty is increased the greater the cumulative reward. When submitting, the user uploads their agent to a private repository, to which the platform has access through an access token. This means that, unlike the competitions in DareFightingICE, the participant's agents or source code are not accessible. Instead, only statistics and video footage of each submission are available. In terms of synchronous tournaments, DIAMBRA hosted its first tournament, RLZ tournament, in 2021. Similarly, the participants' agents and source code are not accessible.

Artificial agents perceive the game state using visual data extracted directly from the game screen, along with some structured features that can be inferred from this visual data. This not only means that the structured features are not strictly necessary to interact in the environments, but also that they do not contain any information hidden from visual perception. The structured data are not extensive, as it does not contain information, such as the players' positions or their current move, which is useful for determining which decisions are appropriate for some state. The intent is to mimic the way human players perceive these games as closely as possible, by focusing on control through visual features. Contrary to DareFightingICE, the structured data are not delayed.

The agent can perform actions in different ways. The platform already discretizes all combinations of movement actions that cover the cardinal and ordinal directions, so other combinations are not necessary. For the attack actions, the platform optionally allows discretizing action combinations, required for some games. Finally, it is possible to perform movement and attack actions either in mutual exclusion or at the same time. This means that the agent can either only perform a movement or an attack action at every frame, or that they can perform both at every frame. The latter is closer to the way humans play these games.

Each game provides a multitude of characters. This is useful for testing generalization, as we can evaluate generalization between games, between characters used by the agent and between characters used by the opponent. In addition, the games provide two modes of play: single-player and multiplayer. The multiplayer mode is similar to how DareFightingICE works, in which two agents/human players fight against each other in isolated games. The single-player mode on the other hand is focused on the agent fighting a sequence of opponents controlled by the AI already implemented in the game. The difficulty rises as the agent advances in the mode. Since agents are matched with opponents of increasing strength, single-player has the potential to be leveraged for curriculum learning.

The platform also provides support for multiple modes of training. Human-in-the-loop training can be performed by taking manual control of the agent and playing against the agent itself during training. It also contains environment wrappers for recording and playing experience, useful for imitation learning.

C. FightLadder

FightLadder is a benchmark focused on competitive multiagent RL [30]. Similarly to DIAMBRA Arena, it emulates various commercial games and provides them as learning environments, which can be interacted with through the Gym API [57]. It supports five commercial games: *Street Fighter II*, *Street Fighter III*, *Fatal Fury 2*, *Mortal Kombat,* and *The King of Fighters '97*. Like the games supported by DIAMBRA Arena, they contain a multitude of characters and game modes. The games support single-player and multiplayer, as well as team mode for *The King of Fighters '97*, in which each player can use multiple characters in a match. Unlike DIAMBRA Arena however, the current selection of games does not cover a large diversity of subgenres, with all games being 2-D nontag fighting games.

The goal of the platform is to provide complex and computationally efficient competitive environments for benchmarking AI based on multiagent RL, which there is a lack of. Due to the lack of unified criteria for evaluating competitive agents, the authors propose four different evaluation criteria. The most basic, and most used in the literature, is using a measure of performance

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图5. DIAMBRA Arena 平台上可用作环境的部分商业游戏。目前的游戏选择丰富多样，涵盖不同的格斗游戏子类型。请注意，平台上提供的漫威对卡普空：超级英雄的碰撞和*X*战警对街头霸王版本支持组队换人格斗，而这些游戏在*Playstation* 主机上发行的版本则不支持。

，并按得分排名。得分通过让所提交的智能体运行一个回合来计算，取决于累积奖励和游戏难度。游戏难度在提交智能体时指定，可选择“简单”“中等”或“困难”；在累积奖励相同的情况下，难度每提高一级，所得分数就会增加。累积奖励越高，各难度对得分的放大效果就越强。提交时，用户将智能体上传至私有代码仓库，平台通过访问令牌访问该仓库。这意味着，与DareFightingICE中的竞赛不同，参赛者的智能体或源代码不对外开放，仅提供每次提交的统计数据和视频录像。在同步锦标赛方面，DIAMBRA于2021年举办了首届锦标赛——RLZ锦标赛。同样，参赛者的智能体和源代码也不对外开放。

人工智能体利用直接从游戏画面中提取的视觉数据，以及可从这些视觉数据中推断出的一些结构化特征来感知游戏状态。这不仅意味着在环境中交互并非严格需要这些结构化特征，也意味着这些特征不包含任何视觉感知无法获取的信息。结构化数据的内容并不丰富，因为其中不包含诸如

玩家的位置或其当前招式等信息，而这些信息有助于确定在某种状态下适合做出哪些决策。其目的是通过侧重于利用视觉特征进行控制，尽可能贴近人类玩家感知这些游戏的方式。与 DareFightingICE 不同，这些结构化数据没有延迟。

智能体可以通过不同方式执行动作。平台已经将涵盖正方向与斜方向的所有移动动作组合离散化，因此无需其他组合。对于攻击动作，平台提供了可选的动作组合离散化功能，某些游戏需要这一功能。最后，移动动作与攻击动作既可以互斥执行，也可以同时执行。这意味着，智能体在每一帧中可以仅执行一个移动动作或一个攻击动作，也可以同时执行这两类动作。后者更接近人类玩这些游戏的方式。

每款游戏都提供众多角色。这有助于测试泛化能力，因为我们可以评估不同游戏之间、智能体所用的不同角色之间，以及对手所用的不同角色之间的泛化能力。此外，这些游戏提供两种游玩模式：单人模式和多人模式。多人模式的运作方式与 DareFightingICE 类似，两个智能体或人类玩家在彼此独立的对局中相互对战。相比之下，单人模式侧重于让智能体与一系列由游戏内置人工智能控制的对手交战。随着智能体在该模式中不断推进，难度也会逐渐上升。由于智能体会匹配到实力不断增强的对手，单人模式具有用于课程学习的潜力。

该平台还支持多种训练模式。可以通过手动控制智能体，并在训练期间与智能体本身对战，来进行人在回路训练。平台还包含用于记录和回放经验的环境封装器，可用于模仿学习。

C. FightLadder

FightLadder 是一个专注于竞争性多智能体强化学习的基准[30]。与 DIAMBRA Arena 类似，它仿真多种商业游戏，并将其作为学习环境，用户可通过 Gym API 与之交互 [57]。它支持五款商业游戏：街头霸王*II*、街头霸王*III*、饿狼传说*2*、真人快打和拳皇*97*。与 DIAMBRA Arena 支持的游戏一样，这些游戏包含众多角色和游戏模式。这些游戏支持单人模式和多人模式，而拳皇*97*还支持团队模式，在该模式下，每位玩家可在一场比赛中使用多个角色。然而，与 DIAMBRA Arena 不同，目前所选游戏并未涵盖丰富多样的子类型，所有游戏均为二维非换人格斗游戏。

该平台旨在提供复杂且计算高效的竞争环境，用于对基于多智能体强化学习的人工智能进行基准测试，而目前此类环境尚显不足。由于缺乏评估竞争性智能体的统一标准，作者提出了四种不同的评估标准。最基本、也是文献中使用最多的标准，是衡量智能体的性能

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against the built-in AI of the respective environment, which commercial games tend to provide. This measure of performance can be, for instance, win rate. Elo ratings were also proposed, which are appropriate when a large pool of opponents is available. These two criteria depend on preexisting agents. The authors also define performance in terms of robustness against exploits by specialized agents. These "exploiter" agents are trained solely to play against the agent to be evaluated, with the objective of overfitting to the agent at the detriment of generalizability. The goal is to measure the transitive strength of the agent's strategy. This criterium contrasts with the previous two in that the measurement is dependent on the training of other agents. This means that the metric is limited in how good of an approximation the exploiter agent is of the best response against the agent, which depends on the algorithm used for the exploiters. Finally, to complement inter-agent evaluation, the authors also propose measuring performance against human players. The advantage of playing manually against the agent is that we can directly influence the game state and it is easier to understand the agent's flaws. However, this analysis is opinionated and heavily dependent of the human players used for evaluation. None of the proposed metrics is perfect on its own, but the combined usage of all of them can bring greater insight into the agent's performance. The platform has features that support evaluation through these metrics.

The platform focuses on visual features, but agents can also use structured data. This structured data can be used to augment the state or to debug the agent. The action space is a simplification of the basic action space where each action is a button press/release. Redundant action combinations are removed, and actions are separated into two sets: movement or defensive actions, and offensive actions. This is similar to how DIAMBRA Arena separates actions between movement and attack actions. Special moves are also discretized, similarly to DareFightingICE.

This is a relatively recent platform that is yet to be explored thoroughly.

VI. Human-Centered Qualities

In the following sections, we summarize the main contributions for each quality of a human-centered agent, gathered from the literature review.

A. Performance

Most works in the development of artificial agents for fighting games tend to focus on performance rather than create agents for play against human players. For two-player zero-sum games, performance is hard to quantify in absolute terms. The evaluation metrics that are used instead are comparative, one example being Elo. These metrics are calculated by having developed agents match against other players, both artificial and human. Table III presents the different kinds of performance evaluation methods employed by various works. Health score is mostly used in the DareFightingICE competition, and is computed as

$$\frac{h_{\text{self}}}{h_{\text{self}} + h_{\text{opp}}} \times a \tag{1}$$

TABLE III
Adoption of Performance Evaluation Methodologies in the Literature

Metric	Applied to	Reference
Win rate	Built-in AI	[70], [71]
	MctsAi	[34], [35], [72], [73]
	Past DareFightingICE competitors	[22], [32], [33], [61], [69], [74]–[79]
	Human players	[25], [30], [73], [80]
Health score	MctsAi	[72]
	Past DareFightingICE competitors	[13], [27], [36], [66], [74], [78], [81]–[86]
Elo rating	Built-in AI	[30]
	Past DareFightingICE competitors	[26]
	Custom agent	[13]
	Own agents	[30]
SDR score	Past DareFightingICE competitors	[24]
Damage dealt	Human players	[14]
	Static opponent	[87]
Average reward	Built-in AI	[67], [80]
	Rule-based AI	[88]
Health difference	MctsAi	[89], [90]
	Past DareFightingICE competitors	[34], [69]
Ranking change	Past DareFightingICE competitors	[62]
K/D ratio	Human players	[5]
Exploitability	Own agents	[30]
Number of hits	Past DareFightingICE competitors	[37]

where a is a multiplicative factor, usually 1000, h_{self} is the health of the agent, and h_{opp} is the health of the opponent. Some works use a slight variation in which the ratio is on damage rather than health, in which case the numerator is on the damage dealt to the opponent, but we treat it as the same metric since they end up representing the same concept. SDR score is computed as

$$\frac{w \times \Delta h}{\Delta t} \tag{2}$$

where w is win rate, Δh is the health difference between both players, and Δt is the elapsed time. Compared with health score, this metric has the advantage that time is incorporated, which values the ability of agents to win quickly. This factor was likely not included in DareFightingICE since the platform already used to host a separate competition for speedrunning, in which agents try beat the opponent as fast as possible.

Although win rate and health score were the most used metrics, there is an opportunity to explore other metrics. For instance, exploitability can be used to test robustness, which can help determine the agent's transitive strength and, therefore, its performance. However, exploitability depends greatly on how the exploiter agents are trained, and due to the metric's recency it likely needs more investigation into reliable methods for producing exploiters. For example, the method in [69] can be used to build exploiter agents, as it is meant to find strategies in the Nash Equilibrium for fighting games. However, it is currently limited in that it does not consider reaction time, which in Section II-A was shown to be an important factor to consider. Elo rating is

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，即其对抗相应环境中内置人工智能时的表现，而商业游戏通常会提供此类内置人工智能。例如，这种性能指标可以是胜率。作者还提出使用Elo等级分，该指标适用于有大量对手可供选择的情况。这两种标准都依赖于已有的智能体。作者还从抵御专用智能体漏洞利用的鲁棒性角度来定义性能。这些“漏洞利用者”智能体仅针对待评估智能体进行对抗训练，目的是以牺牲泛化能力为代价，对该智能体形成过拟合。其目标是衡量该智能体策略的传递性实力。这一标准与前两种标准的区别在于，其衡量结果依赖于其他智能体的训练。这意味着，该指标受到漏洞利用者智能体对最优响应近似程度的限制，而这种近似程度又取决于漏洞利用者所使用的算法。最后，为补充智能体间评估，作者还提出衡量智能体对抗人类玩家时的性能。手动与智能体对战的优势在于，我们可以直接影响游戏状态，也更容易理解智能体的缺陷。然而，这种分析带有主观性，且高度依赖于参与评估的人类玩家。所提出的指标均无法单独做到尽善尽美，但综合使用这些指标可以更深入地了解智能体的性能。该平台提供了支持通过这些指标进行评估的功能。

该平台侧重于视觉特征，但智能体也可以使用结构化数据。这些结构化数据可用于扩充状态或调试智能体。其动作空间是对基础动作空间的简化，后者的每个动作均为一次按键按下／释放。冗余的动作组合被移除，动作被划分为两组：移动或防御动作，以及进攻动作。这与DIAMBRA Arena将动作划分为移动和攻击动作的方式类似。特殊招式也进行了离散化处理，与DareFightingICE类似。

这是一个相对较新的平台，尚有待深入探索。

VI. 以人为中心的特质

在以下各节中，我们将总结文献综述中收集的、针对以人为中心的智能体各项特质的主要贡献。

A. 性能

大多数关于格斗游戏人工智能体开发的研究往往侧重于性能，而非创建用于与人类玩家对战的智能体。对于双人零和游戏，性能难以用绝对数值量化。因此，所采用的评估指标是相对性的，Elo等级分就是其中一例。这些指标通过让开发出的智能体与其他玩家（包括人工智能体和人类玩家）对战来计算。表III列出了各项研究采用的不同类型的性能评估方法。生命值评分主要用于DareFightingICE竞赛，其计算方式为

$$\frac{h_{\text{self}}}{h_{\text{self}} + h_{\text{opp}}} \times a \tag{1}$$

表III 文献中性能评估方法体系的采用情况

Metric	Applied to	Reference
Win rate	Built-in AI	[70], [71]
	MctsAi	[34], [35], [72], [73]
	Past DareFightingICE competitors	[22], [32], [33], [61], [69], [74]–[79]
	Human players	[25], [30], [73], [80]
Health score	MctsAi	[72]
	Past DareFightingICE competitors	[13], [27], [36], [66], [74], [78], [81]–[86]
Elo rating	Built-in AI	[30]
	Past DareFightingICE competitors	[26]
	Custom agent	[13]
	Own agents	[30]
SDR score	Past DareFightingICE competitors	[24]
Damage dealt	Human players	[14]
	Static opponent	[87]
Average reward	Built-in AI	[67], [80]
	Rule-based AI	[88]
Health difference	MctsAi	[89], [90]
	Past DareFightingICE competitors	[34], [69]
Ranking change	Past DareFightingICE competitors	[62]
K/D ratio	Human players	[5]
Exploitability	Own agents	[30]
Number of hits	Past DareFightingICE competitors	[37]

其中，a为乘法因子，通常为1000，h_{self}为智能体的生命值，h_{opp}为对手的生命值。一些研究采用了略有不同的形式，即计算伤害而非生命值的比率，此时分子为对对手造成的伤害，但由于它们最终表达的是同一概念，我们将其视为同一指标。SDR分数的计算方式为

$$\frac{w \times \Delta h}{\Delta t} \tag{2}$$

其中，w为胜率，Δh为双方玩家的生命值差，Δt为经过时间。与生命值评分相比，该指标的优势在于纳入了时间因素，重视智能体快速获胜的能力。DareFightingICE 未纳入这一因素，可能是因为该平台此前已单独举办过速通竞赛，要求智能体尽可能快地击败对手。

尽管胜率和生命值评分是最常用的指标，但仍有探索其他指标的空间。例如，可利用性可用于测试鲁棒性，这有助于确定智能体的传递性实力，进而确定其性能。然而，可利用性在很大程度上取决于利用型智能体的训练方式，而且由于该指标出现时间较短，可能还需要进一步研究生成利用型智能体的可靠方法。例如，[69]中的方法可用于构建利用型智能体，因为该方法旨在寻找格斗游戏中纳什均衡下的策略。然而，该方法目前的局限在于未考虑反应时间，而第II-A节已表明，反应时间是一个需要考虑的重要因素。Elo等级分是

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also a popular metric used in the context of population-based training, and has not been used extensively in fighting games. The advantage of Elo over other metrics, such as win rate, is that it a measure of skill that does not depend on the pairing between players, allowing easier performance comparisons.

Due to the heterogeneity in evaluation methodologies, it is difficult to compare works in terms of performance. Although most works use similar metrics, such as win rate and health score, they evaluate them on different sets of opponents.

To perform the evaluations, the artificial opponents used for testing tend to be the ones already provided in those games, which for commercial games are closed-source. For DareFightingICE the opponents used for comparison tend to be the submissions to past DareFightingICE competitions. In the case of DareFightingICE however using a large set of opponents was shown to be required, and ranking within the competitions is not directly indicative of the competitors' performance [24], [32], [61]. This suggests that there is still much nontransitivity in the strategy space covered by current agents, and as such using many test opponents is required to get a better grasp of agent strength.

Different approaches to improve the performance of artificial agents were used. Stateless agents cannot perform action sequences by themselves on the basic action space, unless the state incorporates historic information [29], [70], [74]. Another possibility is, for the case of special moves, to include the required action sequences and combinations as macro actions in the action space [91]. Since it is possible to perform multiple actions simultaneously, some works tackled actions combinations by treating them as macro actions as well [29], or by having the agent output multiple actions [71].

Some works attempted to improve the agent's ability to perform combos, by including the action history in the state [29] or through planning [33]. However, performing extended combos proves to be difficult, especially in RL as it requires a large amount of exploration [29].

Many works attempted self-play [24], [92], and it was shown to be crucial to improve performance [5], [21], [25]. It was found that naive self-play, i.e., picking the latest agent snapshots, performed worse than randomly picking past agent versions [92]. One possible reason for this result is that picking old versions of the agent provides more diversity, which promotes generalization. It is important to diversify the pool, either by including different opponents [24], using agents with different architectures [21] or including different configurations of the agent, e.g., different reward functions [25], [70]. Firoiu et al. [5] found self-play greatly helped with generalization, which was necessary to perform well against human players. However, the agent could still be exploited and perform odd behavior under certain conditions.

Frame skipping was also employed, in order to remove extreme action granularity and have the agent make decisions at a higher level of temporal abstraction [25]. Another trick employed is to modify the state such that the agent is always facing the same way, i.e., left or right [29]. In these games, it is possible for players to switch sides with the opponent, and since players always face the opponent, they will be facing the opposite side when that switch happens. This brought a performance increase due to the reduced training data imbalance, since we manually perform generalization between states that are semantically the same. This issue can also be tackled at the level of actions, as in [70]. Finally, action masking can be employed [26], [73], since in these games the set of available actions depends on the current state.

Current agents still struggle to maintain good performance under an emulation of reaction time or state/action delay [80], [84]. In [80], it was found that tackling the delay in full, rather than partially, was necessary to obtain good performance. However, the agent was correcting the state/action delay by predicting the future state, and it impacted performance greatly when predicting the full delay, so much so that the agent could not act in real-time.

B. Adaptability

A major problem in creating efficient artificial opponents for human players in fighting games is ensuring a fast enough adaptation to the player's behavior [14]. That is, the artificial agent should change its strategy in response to the opponent's perceived strategy, such that it would be a win-maximizing response. Unless the agents employ a strategy at a Nash Equilibrium, agents cannot be static, or else they are prone to exploitation [32]. As explained in Section II, optimizing play against human players requires adapting to their play style, and not only playing randomly. Adaptation to the player has to be done quickly enough, and not require an extensive number of matches. However, the adaptation should not be too abrupt so as not to cause disorientation to the players, which can occur with ensemble agents [78]. Overall, the literature currently lacks research in this area.

C. Human Likeness

Artificial agents developed for these games tend to behave in a way that human players do not, giving away their artificiality [19]. This is due to the complex decision making and control that human players have, which may even be influenced by emotions, and therefore is hard for agents to emulate. As a result, the players' experience is often diminished due to the lack of believability in the opponent's behavior [18]. This is an imperative concern for fighting games since human players assume that the opponent has exploitable biases in their play style that can be used to their advantage. If it is clear that the opponent is artificial, human players will not think to play in this pattern-seeking way [9], diminishing engagement.

We can consider two approaches to this problem: either the artificial agents are constrained in ways similar to human players which should result in exhibiting similar behavior, or the agents directly attempt to imitate human behavior. Examples of constraint-based approaches are emulation of human reaction time [58], [80] and emulation of pressure or knowledge imperfection [52]. Other than that, current work, including commercial games, tends to focus on the second approach of imitating human behavior.

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基于种群的训练中同样常用的一项指标，但尚未在格斗游戏中得到广泛应用。与胜率等其他指标相比，Elo等级分的优势在于，它是一种不依赖玩家配对情况的技能衡量指标，因此更便于进行性能比较。

由于评估方法存在差异，很难从性能方面对各项研究进行比较。尽管大多数研究使用相似的指标，例如胜率和生命值评分，但它们使用不同的对手集合来评估这些指标。

在进行评估时，用于测试的人工对手往往是这些游戏内置的对手，而商业游戏中的这些对手是闭源的。对于DareFightingICE，用于比较的对手往往是往届DareFightingICE竞赛的参赛作品。然而，研究表明，在DareFightingICE中必须使用较大的对手集合，而且竞赛排名并不能直接反映参赛智能体的性能 [24], [32],[61]。这表明，当前智能体所覆盖的策略空间仍存在大量非传递性，因此需要使用众多测试对手，才能更好地了解智能体实力。

研究采用了不同方法来提升人工智能体的性能。无状态智能体无法自行在基本动作空间中执行动作序列，除非状态中包含历史信息 [29], [70], [74]。对于特殊招式，另一种方法是将所需的动作序列和连招作为宏动作纳入动作空间 [91]。由于可以同时执行多个动作，一些研究也将动作连招视为宏动作来处理 [29], ，或让智能体输出多个动作 [71]。

一些研究尝试通过在状态中纳入动作历史 [29] 或通过规划 [33]来提高智能体执行连招的能力。然而，事实证明，执行长连招十分困难，尤其是在强化学习中，因为这需要大量探索 [29]。

许多研究尝试了自博弈 [24], [92], ，结果表明，自博弈对提升性能至关重要 [5], [21], [25]。研究发现，朴素自博弈，即选取最新的智能体快照，其表现不如随机选取智能体的历史版本 [92]。这一结果的一个可能原因是，选取智能体的旧版本可以提供更多多样性，从而促进泛化。使对手池多样化十分重要，可以通过纳入不同的对手 [24], 、使用不同架构的智能体 [21] ，或纳入智能体的不同配置（例如不同的奖励函数）来实现 [25], [70]。Firoiu 等人 [5] 发现，自博弈对泛化有很大帮助，而泛化是与人类玩家对战时表现出色的必要条件。然而，智能体仍可能被利用，并在某些条件下表现出异常行为。

还采用了跳帧，以避免动作粒度过细，使智能体能够在更高的时间抽象层次上做出决策 [25]。另一种技巧是修改状态，使智能体始终朝向同一方向，即左侧或右侧 [29]。在这些游戏中，玩家可以与对手互换位置，而由于玩家始终面向对手，因此在互换位置时，其朝向也会随之反转。这带来了性能提升，因为我们手动对语义相同的状态进行泛化，从而减少了训练数据的不平衡。这一问题也可以在动作层面解决，如 [70]所示。最后，还可以采用动作掩码 [26], [73], ，因为在这些游戏中，可用动作的集合取决于当前状态。

在仿真反应时间或状态/动作延迟的条件下，当前的智能体仍难以保持良好的性能 [80],[84]。研究 [80], 发现，要获得良好的性能，必须处理全部延迟，而非仅处理部分延迟。然而，该智能体通过预测未来状态来校正状态/动作延迟，而预测整个延迟时段后的状态会极大地影响性能，甚至导致智能体无法实时行动。

B. 适应性

在格斗游戏中，为人类玩家创建高效的人工对手时，一个主要问题是确保其能够足够迅速地适应玩家的行为 [14]。也就是说，人工智能体应根据感知到的对手策略来改变自身策略，从而作出能够最大化获胜概率的应对。除非智能体采用纳什均衡策略，否则其策略不能保持不变，不然就容易被对手利用 [32]。如第II节所述，要优化与人类玩家对战时的表现，就需要适应他们的游戏风格，而不只是随机行动。对玩家的适应必须足够迅速，不能依赖大量对局。然而，适应过程也不应过于突兀，以免使玩家感到无所适从，而集成智能体可能会出现这种情况 [78]。总体而言，现有文献在这一领域仍缺乏研究。

C. 类人性

为这些游戏开发的人工智能体往往会表现出人类玩家不会有的行为，从而暴露其人工智能体的本质 [19]。这是因为人类玩家具有复杂的决策和控制能力，甚至可能受到情绪的影响，因此智能体很难模仿。因此，对手行为缺乏可信性，往往会降低玩家体验 [18]。这对格斗游戏而言是一个亟待解决的问题，因为人类玩家会假定对手的游戏风格中存在可利用的偏差，并借此获得优势。如果对手明显是人工智能体，人类玩家就不会想到以这种寻找行为模式的方式进行游戏 [9]，从而降低参与度。

针对这一问题，我们可以考虑两种方法：一是对人工智能体施加与人类玩家所受限制类似的约束，使其表现出类似的行为；二是让智能体直接尝试模仿人类行为。基于约束的方法包括仿真人类反应时间 [58], [80] ，以及仿真压力或知识的不完备性 [52]。除此之外，当前的研究与实践（包括商业游戏）往往侧重于第二种方法，即模仿人类行为。

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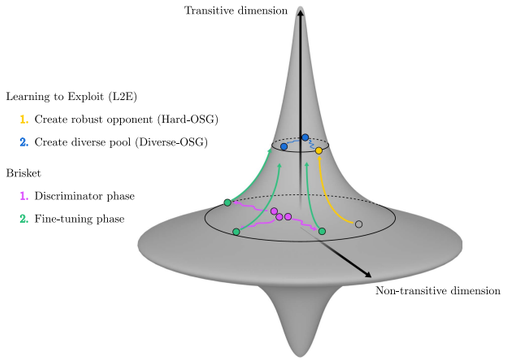


Fig. 6. Strategy space geometry of real world games proposed in [31]. The agent generation strategies in Brisket [28] and L2E [102] are illustrated using this geometry.

Once we have an approach, how do we evaluate "human likeness"? One test seen in some works is the Turing test, alternatively called the "imitation game," in which human evaluators attempt to determine whether a given entity is human or artificial merely by its actions on a particular task [3], [93], [94]. For instance, in one-on-one competitive games, video footage of the artificial agent playing against a human or another artificial opponent is shown to human evaluators. The evaluators have to specify whether each, or one of, the players is human or not. The usefulness of this test is currently a point of controversy [95], but it seems valid since the main objective is to develop an artificial agent behaving similarly to a human, and so the ability of an agent to pass the Turing test directly represents this goal. This is in contrast to the broad original objective of Turing tests, which is to evaluate whether an artificial agent can think intelligently.

Achieving human likeness often conflicts with the pursuit of other qualities. For instance, trying to perform DDA can make the agent have uncompetitive behavior, such as running away when it is performing well, which is unrealistic [96], [97]. This requires the agent to at least avoid strategies that lie below the largest disc in Fig. 6, which correspond to strategies that purposefully attempt to lose. Another example is training using RL. Two works found that RL, while able to improve the agent's performance, usually comes at a cost of introducing artificial behavior [94], [98]. These works introduced a preliminary behavior bias on the agent such that it is prone to producing human-like behavior. This is possible, for example, by using imitation learning with SL, which was shown to be apt for imitating human behavior [20], [98]. Still, the RL algorithms deviated from this bias. Some RL algorithms were found to produce more human-like behavior than others, more specifically policy gradient in comparison with value-based methods, such as Q-learning [5]. One possible factor for this discrepancy is that policy gradient methods directly dictate the policy's stochasticity, whereas in value-based methods it is incorporated manually for the purposes of exploration, and as such contains artificial stochasticity. Still, RL is far from imitation learning in terms of human likeness [20].

D. Difficulty Adjustment

In video games, providing a fair and engaging challenge to players can be done in two main ways: either by having the player choose a difficulty preset according to their perceived skill, or by automatically adjusting game parameters so that adequate difficulty is presented. This automatic process is called DDA, and it is meant to enhance the end users' experience [99]. In this case, we are interested in adjusting the artificial opponent the player is playing against. Usually, estimations of the player's performance are made and parameters are adjusted according to the perceived skill. As such, DDA is also concerned with downgrading the performance of the artificial opponent so that they better match the skill of the human player. In this section, we focus on works that try to implement DDA in fighting games, as that is the method explored in the literature.

Most works that explored DDA in fighting games did so for FightingICE. DDA in fighting games was employed by minimizing absolute health difference between both players [12], [15], [89], [96], [100], excepting one work [101]. Small health differences can suggest that the players are equally matched, but this objective may still produce unnatural or undesired behavior, such as stalling when the agent has a large health lead [15]. Relaxing this objective can produce more proactive agents, rather than reactive, while still keeping the same difficulty adjustment performance [15].

Cherukuri and Glavin [89] followed a different approach, in which the different difficulties correspond to different snapshots of the agent throughout training. During a match, the agent switches between snapshots according to the difference in skill, which is measured as the difference of both players' health. However, this method has the disadvantage that the easiest skill level corresponds to an untrained agent. This means that the agent acts randomly, which is not engaging to play against as it is unpredictable [3].

E. Diversity

Play style can be defined as the way someone or something plays a game, characterized by the strategies that they employ. For games, especially complex ones, a particular skill level is not defined by a single strategy. Fig. 6 illustrates the geometry of the strategy space in real-world games proposed in [31]. The strategy space of real-world games tends to resemble a spinning top, in the sense that the vertical axis encodes the strategy's transitive strength and the width at a particular point in the vertical axis represents the nontransitive dimension, i.e., the breadth of distinct strategies of the respective strength. At the topmost tip of the spinning top we have the Nash equilibria, and the bottom tip contains strategies that purposefully attempt to lose. As such, it is natural that each player has their own play style, defining the way they approach the game, with this being especially true on lower levels of skill. We use this geometry to illustrate our points.

It is worth noting that the geometry of Fig. 6 was proposed considering *symmetric* games, i.e., games in which any two players have access to the same conditions. For example, turn-based games are asymmetric because one player has to start first,

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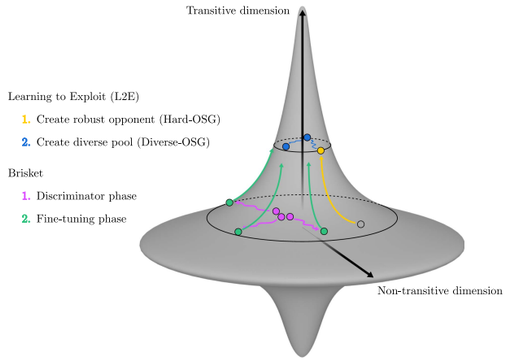


图6。文献中提出的现实世界游戏的策略空间几何结构 [31]。Brisket中的智能体生成策略 [28] 和L2E中的智能体生成策略 [102] 均使用该几何结构加以说明。

有了方法之后，我们该如何评估“类人性”？一些研究采用图灵测试，也称为“模仿游戏”。在该测试中，人类评估者仅根据某个实体执行特定任务时的行为，尝试判断其是人类还是人工智能体 [3], [93], [94]。例如，在一对一竞技游戏中，会向人类评估者展示人工智能体与人类或另一人工智能对手对战的视频片段。评估者需要指出每位玩家或其中一位玩家是否为人类。目前，这种测试的实用性仍存在争议 [95],，但它似乎是合理的，因为主要目标是开发行为与人类相似的人工智能体，因此，智能体能否通过图灵测试直接体现了这一目标是否达成。这与图灵测试最初的宽泛目标有所不同，后者旨在评估人工智能体是否能够进行智能思考。

实现类人性往往与追求其他特性相冲突。例如，尝试进行动态难度调整可能会使智能体表现出缺乏竞争性的行为，例如在表现良好时逃跑，这并不真实 [96],[97]。这要求智能体至少避免采用图6中最大圆盘下方的策略，这些策略对应的是刻意求败的策略。另一个例子是使用强化学习进行训练。两项研究发现，强化学习虽然能够提升智能体的性能，但通常也会以引入不自然行为为代价 [94], [98]。这些研究为智能体引入了初始行为偏置，使其倾向于产生类人行为。例如，可以通过结合监督学习的模仿学习来实现这一点，研究表明，这种方法适合模仿人类行为 [20], [98]。然而，强化学习算法仍然偏离了这一偏置。研究发现，一些强化学习算法比其他算法更能产生类人行为，具体而言，策略梯度比Q学习等基于价值的方法更能产生类人行为 [5]。造成这种差异的一个可能因素是，策略梯度方法直接决定策略随机性，而在基于价值的方法中，随机性是为了探索而人为引入的，因此包含人为随机性。尽管如此，在类人性方面，强化学习仍远不及模仿学习 [20]。

D. 难度调整

在电子游戏中，为玩家提供公平且引人入胜的挑战主要有两种方式：一是让玩家根据自己感知的技能水平选择难度预设，二是自动调整游戏参数以提供适当的难度。这一自动过程称为动态难度调整，旨在改善最终用户的游戏体验 [99]。在这里，我们关注的是调整与玩家对战的人工智能对手。通常，系统会估计玩家的表现，并根据感知到的技能水平调整参数。因此，动态难度调整也涉及降低人工智能对手的性能，使其更好地匹配人类玩家的技能水平。本节重点介绍尝试在格斗游戏中实现动态难度调整的研究，因为这是现有文献所探索的方法。

大多数探索格斗游戏中动态难度调整的研究都是针对FightingICE开展的。格斗游戏中的动态难度调整通过最小化双方玩家之间的生命值绝对差来实现 [12], [15], [89], [96], [100],，仅有一项研究例外 [101]。较小的生命值差可能表明玩家实力相当，但这一目标仍可能导致不自然或不符合预期的行为，例如智能体在拥有较大生命值优势时采取拖延行为 [15]。放宽这一目标可以产生更具主动性而非被动反应的智能体，同时仍保持相同的难度调整性能 [15]。

Cherukuri和Glavin [89] 采用了另一种方法，其中不同难度对应于智能体在训练过程中不同阶段的快照。在对局期间，智能体根据技能差距在快照之间切换，该差距以双方玩家的生命值差来衡量。然而，这种方法的缺点在于，最低技能水平对应的是未经训练的智能体。这意味着智能体会随机行动，其行为难以预测，因此与之对战缺乏吸引力 [3]。

E. 多样性

游戏风格可定义为某人或某个实体玩游戏的方式，其特征体现在所采用的策略上。对于游戏，尤其是复杂游戏而言，特定的技能水平并非由单一策略所定义。图6 展示了 [31]中提出的现实世界游戏策略空间的几何形态。现实世界游戏的策略空间往往类似于陀螺：纵轴表示策略的传递性实力，而纵轴上某一点处的宽度则表示非传递性维度，即相应实力下不同策略的广度。陀螺的最顶端是纳什均衡，最底端则包含刻意求败的策略。因此，每位玩家自然都有自己的游戏风格，决定了他们参与游戏的方式，在技能水平较低时尤其如此。我们使用这一几何形态来阐明我们的观点。

值得注意的是，图6 中的几何结构是在考虑对称博弈的情况下提出的，即任意两名玩家都具备相同条件的博弈。例如，回合制游戏属于非对称博弈，因为一名玩家必须先行动，

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and the other can only act afterward. Fighting games are also asymmetric unless both players are utilizing the same character. The authors include asymmetric games in their analysis by considering each player to play once in the other player's conditions, which in the case of fighting games means two matches are performed where each player plays once with their character and then again with their opponent's character.

This is an important formalism especially when considering population-based training methods that are often used to achieve high performance in competitive environments by artificial agents. In these competitive environments, in which the artificial agent is playing against other adversaries, we need to supply a good opponent pool such that we improve the agent by increasing their transitive strength. However, we may find that when using an opponent pool populated by naive self-play, for instance, the agent is just navigating along the nontransitive dimension, learning policies that merely form strategy cycles during training and not improving overall. Different self-play strategies have been adopted to attempt to avoid this, such as populating the opponent pool with "exploiter" agents that are focused on exploiting the main agent's weaknesses rather than trying to be robust against many opponents [103]. Essentially, we want to avoid traversing the nontransitive dimension and traverse the transitive one as much as possible. One theoretical result is that, if the population size is representative enough of the number of strategies of a given transitive strength, then we can guarantee improvement in terms of transitive strength. Moreover, we can note that nontransitive complexity lowers as the agent gets stronger, assuming the agent is beyond the largest disc of the spinning top. As such, there is interest in properly initializing the opponent pool so that the agent gets to a higher transitive strength as quickly as possible, e.g., by initializing the population with highly skilled opponents. In the FightingICE competition, for instance, since 2020 it has been frequent for competitors to use submissions from past years as training opponents in self-play [24].

We can additionally induce that, to create engaging play in fighting games, it is not enough to just create a single opponent. There is interest in having a population of agents, each exhibiting diverse behaviors that better represent the game's strategy space. This way, human players have to figure out and better adapt to various play styles of the artificial opponents, at any skill level. In addition, given the spinning top structure, some skill levels exhibit a greater set of strategies, so the population size can vary among skill levels.

Not many works in the literature explore this area. Of note, Halina and Guzdial [28] developed Brisket, a method for creating a population of agents with diverse behaviors for FightingICE. Training is separated into two phases, where in the first phase agents merely learn to exhibit diverse behaviors between each other, whether they be good or bad for performance at the game. Then, with behavioral bias set from the previous phase, the second phase focuses on learning how to play the game, using DQN with reduced learning rate to avoid radical changes in behavior. These two phases can be thought of as the first one focusing on traversing the game's spinning top space in the nontransitive dimension, while the second phase focuses on traversing the transitive one. This behavior diversity is achieved using an adaptation of the diversity is all you need (DIAYN) method [104], which is a method for allowing an agent to learn a fixed set of skills that allow it to traverse distinct portions of the state space. In Brisket, instead of having diversity between skills in a single agent, we have diversity between different agents in an agent pool. As such, whereas DIAYN is focused on within-agent diversity, Brisket is focused on between-agent diversity. The developed agents were diverse, contrasting with the human-authored baselines developed through reward shaping, which converged to the same local maxima. At the same time, all agents exhibited the same performance, which was also superior to the baselines' performance.

Fig. 6 illustrates the Brisket population generation algorithm, as well as the learning to exploit (L2E) framework for comparison [102]. Although not applied to fighting games, the L2E work proposes an opponent strategy generation (OSG) algorithm similar to Brisket's, specifically for the generation of diverse training opponents. The algorithm is separated into two phases, but with inverted goals: in the first phase (Hard-OSG), a single randomly initialized opponent is optimized to be robust against the current agent; in the next phase (Diverse-OSG), a diverse opponent pool is iteratively created by creating opponents similarly to the first phase, but with the additional constraint of being dissimilar to all opponents in the pool so far. Diversity is enforced by ensuring opponents have behavior that produces distinct trajectories during gameplay. Through ablation tests, the authors found that utilizing both optimization phases (optimizing for robustness and diversity) helps increase the agent's expected return, with diversity being a large contributing factor. This work highlights the need to have not only a robust but also a diverse set of opponents during training, especially important when employing self-play. We find this work worth mentioning as a contrast to Brisket, and highlight how population generation for fighting games can still be explored further as it is not a clear science.

F. Generalizability

Considering the variety of characters present in these games, it is desirable that agents have a degree of generalization. Agents should be able to transfer skills not only between playable characters but also between opponent characters. This can be performed through transfer learning. Transfer learning denotes the process of an ML model, trained for a given task, to be trained for a different task such that the bias brought by the first task improves learning and performance on the second task. This is desirable when the two tasks are only slightly different and makes training much more efficient. In fighting games, we can think of transfer learning as the ability for an agent, trained on playing as a specific character, to be able to play as another character, but without requiring the onerous training as was needed for the first character since some skills should be transferable. Or, similarly, an agent trained to play *against* a character or opponent being able to efficiently learn how to play against another one. This is especially pertinent when games have a large character roster [21]. As such, transfer

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而另一名玩家只能随后行动。格斗游戏也属于非对称博弈，除非双方玩家使用相同的角色。作者通过让每名玩家在对方的条件下各进行一次游戏，将非对称博弈纳入分析。对于格斗游戏而言，这意味着进行两场比赛，每名玩家先使用自己的角色进行一场比赛，再使用对手的角色进行一场比赛。

这是一种重要的形式化描述，尤其是在考虑基于种群的训练方法时，人工智能体常使用这类方法在竞争环境中实现高性能。在这些竞争环境中，人工智能体需要与其他对手进行对抗，因此我们需要提供一个良好的对手池，通过提升智能体的传递性实力来改进智能体。然而，我们可能会发现，例如，当使用由朴素自博弈生成的对手所构成的对手池时，智能体只是在非传递性维度上游走，在训练期间学到的策略仅仅构成策略循环，整体上并没有得到改进。为尝试避免这一问题，研究者采用了不同的自博弈策略，例如向对手池中加入"利用者"智能体，这些智能体专注于利用主要智能体的弱点，而非力求在面对众多对手时保持鲁棒性 [103]。本质上，我们希望避免沿非传递性维度移动，而是尽可能沿传递性维度移动。一项理论结果表明，如果种群规模足以代表给定传递性实力下的策略数量，那么就能保证传递性实力得到提升。此外，我们可以注意到，假设智能体已超越陀螺的最大圆盘所在位置，那么随着智能体实力增强，非传递性复杂度会降低。因此，如何恰当地初始化对手池，使智能体尽快达到更高的传递性实力，是一个值得关注的问题，例如可以用高水平对手来初始化种群。以FightingICE竞赛为例，自2020年以来，参赛者经常将往年的参赛程序用作自博弈中的训练对手 [24]。

我们还可以进一步推断，要在格斗游戏中创造引人入胜的对战体验，仅创建单个对手是不够的。值得考虑的是构建一个智能体群体，让各智能体展现出多样化的行为，从而更好地体现游戏的策略空间。这样，无论处于何种技能水平，人类玩家都必须摸清人工对手的各种游戏风格，并更好地加以适应。此外，鉴于陀螺结构，某些技能水平所涵盖的策略集合更大，因此不同技能水平下的种群规模可以有所不同。

现有文献中探索这一领域的工作并不多。值得注意的是，Halina和Guzdial [28] 开发了Brisket，这是一种为FightingICE创建具有多样化行为的智能体群体的方法。训练分为两个阶段：在第一阶段，智能体仅学习表现出彼此不同的行为，而不考虑这些行为对游戏性能的影响是好是坏。随后，在依据上一阶段确定行为偏置后，第二阶段专注于学习如何进行游戏，使用学习率较低的深度Q网络，以避免行为发生剧烈变化。这两个阶段可以理解为：第一阶段侧重于沿非传递性维度遍历游戏的陀螺空间，而第二阶段则侧重于沿传递性维度进行遍历。这种行为多样性是通过对多样性即你所需（DIAYN）方法进行适应性调整来实现的。[104], 该方法使智能体能够学习一组固定的技能，从而遍历状态空间中的不同部分。在Brisket中，多样性并非体现在单个智能体的不同技能之间，而是体现在智能体池中的不同智能体之间。因此，DIAYN侧重于智能体内多样性，而Brisket侧重于智能体间多样性。所开发的智能体表现出多样性，与通过收益塑造开发的人工设计的基线形成对比，后者均收敛至同一局部极大值。同时，所有智能体均表现出相同的性能，且优于基线的性能。

图 6 展示了 Brisket 种群生成算法，以及用于比较的学习利用（L2E）框架 [102]。尽管并未应用于格斗游戏，L2E 研究提出了一种与 Brisket 类似的对手策略生成（OSG）算法，专门用于生成多样化的训练对手。该算法分为两个阶段，但目标与 Brisket 的顺序相反：在第一阶段（Hard-OSG），对单个随机初始化的对手进行优化，使其在面对当前智能体时具备鲁棒性；在下一阶段（Diverse-OSG），以类似于第一阶段的方式创建对手，并通过迭代构建一个多样化的对手池，但额外要求新对手与池中已有的所有对手均不相似。通过确保对手的行为在游戏玩法过程中产生不同的轨迹，来保证多样性。通过消融测试，作者发现，同时采用这两个优化阶段（针对鲁棒性和多样性进行优化）有助于提高智能体的期望回报，其中多样性发挥了重要作用。这项研究强调，训练期间不仅需要一组具有鲁棒性的对手，还需要这组对手具有多样性，这在采用自博弈时尤为重要。我们认为，值得将这项研究作为 Brisket 的对照加以介绍，同时也要强调，格斗游戏的种群生成尚未形成明确的科学方法，仍有进一步探索的空间。

F. 泛化能力

考虑到这些游戏中的角色多种多样，智能体最好具备一定的泛化能力。智能体不仅应能够在不同可玩角色之间迁移技能，还应能够在不同对手角色之间迁移技能。这可以通过迁移学习来实现。迁移学习是指将针对某一任务训练的机器学习模型用于另一任务的训练，使第一个任务所带来的偏置能够改善第二个任务的学习效果和性能。当两个任务仅有细微差异时，这种方法尤为适用，并能显著提高训练效率。在格斗游戏中，我们可以将迁移学习理解为：一个经过训练、能够操控特定角色的智能体，也能操控另一个角色，但由于部分技能应当可以迁移，因此无须像训练第一个角色时那样进行繁重的训练。类似地，一个经过训练、能够对抗某个角色或对手的智能体，也能够高效地学会如何对抗另一个角色或对手。当游戏拥有庞大的角色阵容时，这一点尤为重要 [21]。因此，迁移

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TABLE IV
WORKS RELATED WITH EACH PAIR OF AGENT-DEVELOPMENT TECHNIQUE AND QUALITY OF A HUMAN-CENTERED AGENT

Technique	Performance	Adaptability	Human likeness	Difficulty adjustment	Diversity	Generalizability
Rule-based	[13], [14], [78], [83], [84], [115]	[13], [14], [78]	[14], [52], [115], [116]	[52]	[52]	—
Mimicry	[14], [106]	[14], [41]	[14], [41], [50]	—	—	—
SL	[34], [36]–[38], [77], [81], [82], [84], [109]	—	[3], [20], [116]	[23], [101]	—	—
Planning	[4], [27], [32]–[36], [61], [62], [66], [73], [77], [79], [81], [85], [86], [90], [109], [110], [115]	[33]	[4], [96], [115]	[15], [96], [101], [112]	[111]	—
Evolutionary algorithms	[22], [33]–[38], [40], [66], [75], [83], [90], [110]	[33]	[19], [114]	[113]	[114]	—
Reinforcement learning	[5], [12], [22], [24]–[30], [32], [34], [37], [66], [67], [69]–[74], [76], [80], [82], [87]–[89], [91], [94], [117], [118]	[16], [17], [29], [105]	[5], [20], [21], [24], [29], [80], [94]	[12], [21], [23], [89], [97], [100], [112], [113]	[28], [30], [70], [94]	[5], [21], [29], [30], [92], [97]

learning has the potential to greatly reduce the time taken to train artificial opponents. To facilitate development and training, we may also be interested in having agents that are generalizable across games.

Transfer learning between player characters [5], [21] and opponent characters [29] was shown to reduce training time greatly. Curiously, in [5], the learning efficiency for each character largely corroborated the human players' perception of learning difficulty for those characters.

In terms of performance, Kim et al. [105] found that representation transfer can make adaptation to different environments easier, but can impact performance in the long-term. That is, the agent can be biased during the adaptation.

To improve generalizability, Pathak et al. [21] removed identifiers, such as the move each player is performing, from the state features. As such, only quantitative information was kept, such as the players' position and health. Although this reduced performance for specific matchups, excluding it allowed agents to exhibit more consistent behavior throughout matchups. This is because identifiers are too discriminative of each situation, and can make it difficult to generalize.

VII. DISCUSSION

In this section, we summarize the work on the main method groups identified during our survey. We present the general conclusions taken for each method group, with respect to the qualities of a human-centered agent we described previously. Table IV presents the works we identified that relate to each method-quality pair, condensing 77 works in total. We can notice at a glance the lack of study for some method-quality pairs, such as rule-based methods for generalizability.

Beyond the rule-based, planning and RL methods prominent in the DareFightingICE competitions, we identified three others: mimicry methods aimed at imitating the behavior of a player, SL methods used either for building a policy or for opponent modeling, and evolutionary algorithms, used for multiple purposes.

We found rule-based agents mostly in the DareFightingICE competitions; not many works in the literature exist exploring this methodology. The works we studied used either dynamic scripting [13], [14] or a mixture of experts [78], with one of them winning first place in the FightingICE competition [13]. In terms of adaptability, the mixture of experts method can have trouble adapting to the opponent since the internal agents are usually made static [78]. Dynamic scripting on the other hand has found more success, with the agent being capable of adapting within a round [13]. In dynamic scripting, the agent's behavior script is made specifically for its parts to be adjusted, and this increased flexibility allows for better adaptability.

Works based on mimicry are focused on imitating another player's behavior rather than performance. Methods include case-based reasoning [14], [41], [50] and unsupervised learning [106]. At their essence, these methods use a dataset of human play data to determine which actions to perform at the current state. One work decided to implement "skepticism"[14], which allows the agent to opt for not imitating actions, which it deems to not be useful to achieve its goal. Skepticism not only gives the agent the chance to surpass the imitated player in terms of performance, but it was also found to increase human likeness. This is a concept that can be useful to apply for other methods. The agent's actions need to span the whole action set, including the action of doing nothing, as waiting is valid human behavior [50]. For adaptability, case-based reasoning needs to consider the most recent cases and discard or give less weight to older ones [41].

SL is usually used for imitation learning [20], [37], [38] or opponent modeling [34], [36], [107], [108], [109]. In terms of performance, imitation learning is limited since the agent can

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表IV
W各智能体开发技术与以人为中心的智能体质量属性组合的相关研究 T

Technique	Performance	Adaptability	Human likeness	Difficulty adjustment	Diversity	Generalizability
Rule-based	[13], [14], [78], [83], [84], [115]	[13], [14], [78]	[14], [52], [115], [116]	[52]	[52]	—
Mimicry	[14], [106]	[14], [41]	[14], [41], [50]	—	—	—
SL	[34], [36]–[38], [77], [81], [82], [84], [109]	—	[3], [20], [116]	[23], [101]	—	—
Planning	[4], [27], [32]–[36], [61], [62], [66], [73], [77], [79], [81], [85], [86], [90], [109], [110], [115]	[33]	[4], [96], [115]	[15], [96], [101], [112]	[111]	—
Evolutionary algorithms	[22], [33]–[38], [40], [66], [75], [83], [90], [110]	[33]	[19], [114]	[113]	[114]	—
Reinforcement learning	[5], [12], [22], [24]–[30], [32], [34], [37], [66], [67], [69]–[74], [76], [80], [82], [87]–[89], [91], [94], [117], [118]	[16], [17], [29], [105]	[5], [20], [21], [24], [29], [80], [94]	[12], [21], [23], [89], [97], [100], [112], [113]	[28], [30], [70], [94]	[5], [21], [29], [30], [92], [97]

学习有望大幅缩短训练人工对手所需的时间。为了便于开发和训练，我们也可能希望智能体具备跨游戏的泛化能力。

玩家角色之间的迁移学习 [5], [21] 以及对手角色之间的迁移学习 [29] 已被证明能够大幅缩短训练时间。有趣的是，在 [5], 中，各角色的学习效率在很大程度上印证了人类玩家对这些角色学习难度的感知。

在性能方面，Kim等人 [105] 发现，表征迁移能够使适应不同环境变得更容易，但可能影响长期性能。也就是说，智能体在适应过程中可能产生偏置。

为提高泛化能力，Pathak等人 [21] 从状态特征中移除了标识符，例如各玩家正在执行的招式。因此，仅保留了玩家位置和生命值等定量信息。尽管这降低了特定对阵中的性能，但排除这些标识符后，智能体在不同对阵中表现出了更一致的行为。这是因为标识符对各类情境的区分过于鲜明，可能使泛化变得困难。

VII. 讨论

本节总结了本次综述中确定的主要方法类别的相关工作。针对前文所述的以人为中心的智能体应具备的各项特质，我们给出了对各类方法的总体结论。表IV 列出了我们找到的与各个“方法—特质”组合相关的研究，共汇总了77项工作。我们可以一目了然地看出，某些“方法—特质”组合尚缺乏研究，例如基于规则的方法在泛化能力方面的研究。

除了在DareFightingICE竞赛中占据主导地位的基于规则、规划和强化学习方法外，我们还识别出另外三类方法：旨在模仿玩家行为的模仿方法，用于构建策略或对手建模的监督学习方法，以及用于多种用途的进化算法。

我们发现，基于规则的智能体主要出现在DareFightingICE竞赛中；现有文献中探索这类方法的研究并不多。我们考察的研究采用了动态脚本 [13], [14] 或混合专家 [78], ，其中一项研究在FightingICE竞赛中获得了第一名 [13]。在适应性方面，混合专家方法可能难以适应对手，因为其内部智能体通常被设定为静态的 [78]。相比之下，动态脚本取得了更好的成效，智能体能够在一局之内完成适应 [13]。在动态脚本中，智能体的行为脚本经过专门设计，使其各个部分均可调整，而这种更高的灵活性带来了更好的适应性。

基于模仿的研究侧重于模仿另一名玩家的行为，而非其性能。所采用的方法包括基于案例的推理 [14], [41], [50] 和无监督学习 [106]。这些方法的本质是利用人类游戏数据集来确定在当前状态下应执行哪些动作。有一项研究引入了“怀疑机制”[14]，使智能体能够选择不模仿其认为无助于实现目标的动作。怀疑机制不仅使智能体有机会在性能上超越被模仿的玩家，而且研究发现，它还能提升类人性。这一概念也可用于其他方法。智能体的动作需要覆盖整个动作集，包括不采取任何行动，因为等待也是合理的人类行为 [50]。为实现适应性，基于案例的推理需要考虑最新的案例，并丢弃较早的案例或降低其权重 [41]。

监督学习通常用于模仿学习 [20], [37], [38] 或对手建模 [34], [36], [107], [108], [109]。在性能方面，模仿学习存在局限，因为智能体

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only, at best, match the performance of the player it is imitating. Most of the surveyed applications of SL either used offline learning [20], [37], [38], [107], [108] or performed learning but only between game rounds [34]. To perform difficulty adjustment, SL has been used to classify the affective state of the player to support a MCTS agent targeting that state [101]. Imitation learning with SL proves successful in achieving human-like behavior, when compared with RL [20].

Planning is usually not performed in isolation. It incorporates other methods, such as opponent modeling or RL, in order to improve the planning process, and has faced success when evaluated with opponents from the DareFightingICE competitions. For instance, the reference MctsAi in DareFightingICE assumes a random opponent during simulation [86], so some works utilized opponent modeling to make simulations more accurate [27], [34], [36], [61], [85]. On the other hand, the action set considered during planning can also be shrunk by only considering the best candidate actions, through RL [27], [32] or evolutionary algorithms [35], [110]. Planning has also been augmented through selection of hyperparameters with evolutionary algorithms [90], by including rule-based behavior to handle specific situations [62], [85], and approximation of the state evaluation function [77]. In terms of adaptation, the surveyed planning methods are dependent on the augmentation method used, e.g., the efficiency with which the opponent model adapts to the opponent [34]. Pure planning methods have still not explored believability extensively, and are yet to produce human-like behavior [4]. The planning goal can be used to define different play styles, e.g., by considering player distance [111], and for difficulty adjustment, by targeting similar health values [15], [96], [112] or affective states [101].

Evolutionary algorithms, such as genetic algorithms, are used for multiple purposes. They are used to augment planning [33], [34], [35], [66], [90], build neural networks [22], [36], [37], [38] and find combos [19], [40], [110]. Given their nature, evolutionary algorithms are not fit for adaptation unless used alongside other methods, since the iterative evolutionary process to build candidate individuals takes time. Even then, the approaches that combined evolutionary algorithms with other methods exhibited either no adaptation at all due to the generated policy or dataset being static [37], [66], [110], [113], or the methods did not learn fully online [34]. One work attempted to build a diverse population, but the agents had poor performance [114].

RL has been widely explored in the literature. It can still be difficult to apply in practice, with mixed success in the DareFightingICE competitions and against human players. Separating the reward function into subgoals can yield more stable performance [74], and using completely separate predictors for those subgoals can be even more effective [72]. To match the skill of human players, self-play was found to be an important training method for RL, either using only training snapshots [5], [25], [92] or including premade agents [24]. The methods applied in practice are sample inefficient [29], [32], [94], taking hours to train, which limits adaptability. Still, representation transfer has shown to help increase training speed [105]. Human likeness is usually implemented through constraints on the agent's behavior, either by imposing an observation or action delay emulating the reaction time of human beings [21], [24], [80], or by restricting the agent's exploration process to avoid weird behavior [4], [94]. To handle reaction time delay, prediction of the current frame for observation delay [24], necessary for DareFightingICE, and of the future frame for action delay [21], [80] have been studied. Reward shaping has been used both to restrict the agent's behavior to be more human-like [29], and to build diverse agents [94]. Diversity can also be enforced within a population of agents through a discriminatory procedure [28]. For difficulty adjustment, we can save various snapshots of the agent throughout training and associate training time with agent skill [89]. Alternatively, we can have the agent purposefully choose suboptimal actions [100] or use custom reward so as to minimize the absolute health difference between players [12]. Finally, generalizability has been studied for RL only, and it was shown that transfer learning can be applied both between playable characters [5], [21] and opponent characters [29]. However, having an agent learn multiple playable characters at the same time is still difficult, and it is recommended to either have each agent focus on a particular character [5] or drop discrete information, such as identifiers, in the state [21].

VIII. Conclusion

In this work, we motivate the existence of human-centered agents for fighting games. Human-centered agents are artificial agents built with the human-AI gameplay experience in mind, providing an experience similar to playing with human players while being adjusted to the player's profile. We surveyed the literature for agent development methodologies for fighting games, as well as learning environments on which those agents can be developed, and presented their contributions with respect to each quality of a human-centered agent.

Most commercial fighting games either do not reveal their approach to agent development, or employ methods based on imitation of other players. Some works in the literature develop agents within commercial fighting games, often through emulation. This process is simplified in DIAMBRA Arena and Fight-Ladder, platforms containing various learning environments for RL-based AI, with each environment emulating a commercial fighting game. Another learning environment is DareFightingICE, which offers a game purposefully made for research and competition of AI.

Despite their straightforward approach, rule-based agents are able to be performant in some situations where automated learning approaches such as SL and RL fail. However, they are also less generally applicable since they have to be manually built. RL methods for fighting games are shown to be successful, even able to beat expert human opponents, but can still show odd behavior sometimes. Evolutionary and planning algorithms are often mixed with other methods, such as RL, to further improve performance.

There still has not been extensive work on the efficiency of adaptation to varying opponents in fighting games. Works evaluate whether adaptation occurs at all and if it leads to good performance, but do not focus on its efficiency.

Exhibition of human-like behavior is hard to compare due to the subjective nature of the evaluation. Nevertheless, we can note the success of imitation learning and case-based reasoning,

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充其量只能达到其所模仿玩家的性能水平。在所综述的监督学习应用中，大多数要么采用离线学习 [20], [37], [38], [107], [108] ，要么仅在游戏回合之间进行学习 [34]。为实现难度调整，监督学习被用于对玩家的情感状态进行分类，以支持针对该状态的蒙特卡洛树搜索智能体 [101]。与强化学习相比，采用监督学习的模仿学习已被证明能够成功实现类人行为 [20]。

规划通常并非独立进行。它会结合其他方法，例如对手建模或强化学习，以改进规划过程，并且在与 DareFightingICE 竞赛中的对手进行对战评估时取得了成功。例如，DareFightingICE 中的基准智能体 MctsAi 在仿真期间假定对手随机行动 [86], ，因此，一些研究利用对手建模来提高仿真的准确性 [27], [34], [36], [61], [85]。另一方面，也可以通过强化学习，仅考虑最佳候选动作，从而缩小规划期间所考虑的动作集 [27],[32] ，或通过进化算法实现这一点 [35], [110]。此外，还可以通过利用进化算法选择超参数 [90], 、引入基于规则的行为来处理特定情形 [62], [85], 以及近似状态评估函数来增强规划 [77]。在适应方面，所综述的规划方法依赖于所采用的增强方法，例如，对手模型适应对手的效率 [34]。纯规划方法尚未广泛探索可信性，也尚未产生类人行为 [4]。规划目标可用于定义不同的游戏风格，例如通过考虑玩家距离 [111], ；也可用于难度调整，例如以相近的生命值 [15], [96], [112] 或情感状态为目标 [101]。

进化算法（如遗传算法）有多种用途，可用于增强规划 [33],[34], [35], [66], [90], 、构建神经网络 [22], [36], [37], [38] 和寻找连招 [19], [40], [110]。由于构建候选个体的迭代进化过程需要时间，进化算法本身并不适合用于适应，除非与其他方法结合使用。即便如此，将进化算法与其他方法结合的方案，要么因生成的策略或数据集是静态的而完全不具备适应能力 [37], [66], [110], [113], ，要么无法完全在线学习 [34]。有一项研究尝试构建多样化种群，但智能体的性能较差 [114]。

强化学习已在文献中得到广泛探索。但在实践中应用它仍可能存在困难，其在DareFightingICE竞赛中以及与人类玩家对战时的表现有好有坏。将奖励函数拆分为子目标可以获得更稳定的性能 [74]，而为这些子目标使用完全独立的预测器则可能更加有效 [72]。研究发现，要达到人类玩家的技能水平，自博弈是强化学习的一种重要训练方法，无论是仅使用训练快照 [5], [25],[92] ，还是加入预制智能体 [24]。实际应用的方法样本效率较低 [29], [32], [94], ，需要数小时训练，这限制了适应性。不过，表征迁移已被证明有助于提高训练速度 [105]。类人性通常通过约束智能体的行为来实现，例如施加观测或动作延迟以模拟人类的反应时间 [21], [24], [80], 或者限制智能体的探索过程以避免异常行为 [4], [94]。为应对反应时间延迟，研究者已研究了针对 DareFightingICE所必需的观测延迟进行当前帧预测 [24], ，以及针对动作延迟进行未来帧预测 [21],[80] 。奖励塑形既被用于约束智能体的行为，使其更具类人性 [29], ，也被用于构建多样化的智能体 [94]。还可以通过判别过程在智能体群体内强制实现多样性 [28]。对于难度调整，我们可以在整个训练过程中保存智能体的多个快照，并将训练时间与智能体技能关联起来 [89]。另一种方法是让智能体有意选择次优动作 [100] ，或使用自定义奖励来尽可能缩小玩家之间的生命值绝对差 [12]。最后，目前仅针对强化学习研究了泛化能力，结果表明，迁移学习既可以应用于不同可玩角色之间 [5], [21] ，也可以应用于不同对手角色之间 [29]。然而，让一个智能体同时学习多个可玩角色仍然很困难，因此建议让每个智能体专注于特定角色 [5] ，或从状态中去除标识符等离散信息 [21]。

VIII. 结论

本文阐述了在格斗游戏中引入以人为中心的智能体的必要性。以人为中心的智能体是在人机游戏体验的考量下构建的人工智能体，能够根据玩家画像进行调整，同时提供类似于与人类玩家对战的体验。我们综述了有关格斗游戏智能体开发方法以及可用于开发这些智能体的学习环境的文献，并从以人为中心的智能体的各项特质出发，介绍了这些研究的贡献。

大多数商业格斗游戏要么不公开其智能体开发方法，要么采用基于模仿其他玩家的方法。文献中的一些研究在商业格斗游戏中开发智能体，通常通过仿真来实现。DIAMBRA Arena 和 Fight-Ladder 简化了这一过程；这两个平台包含多种面向基于强化学习的人工智能的学习环境，每个环境都仿真一款商业格斗游戏。另一个学习环境是 DareFightingICE，它提供了一款专为人工智能研究和竞赛而设计的游戏。

尽管方法简单直接，基于规则的智能体仍能在监督学习和强化学习等自动化学习方法失效的某些情境下取得良好性能。然而，由于必须手动构建，这类智能体的通用性也较低。用于格斗游戏的强化学习方法已被证明能够取得成功，甚至能够击败人类专家对手，但有时仍会表现出异常行为。进化算法与规划算法通常与强化学习等其他方法结合使用，以进一步提升性能。

目前，针对格斗游戏中适应不同对手的效率，尚未开展广泛研究。现有研究评估是否发生了适应，以及适应是否带来了良好性能，但并未重点关注适应的效率。

由于评估具有主观性，类人行为的表现难以比较。不过，我们可以看到模仿学习和基于案例的推理取得了成功，

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which are even the methods most opted for in commercial games. RL on the other hand tends to produce artificial behavior, even when mixing it with imitation learning. Although RL had less success, we can note hierarchical reinforcement learning and policy gradient methods produced more human-like behavior than other approaches within this paradigm, such as value-based methods. Still, many works on human likeness do not take into account the fairness of the agent, such as emulation of reaction time. The influence of a reaction time constraint on human-like behavior is yet to be determined.

There are also some key insights for the development of engaging artificial agents. For one, implementing various difficulty levels through versions of the agent at different stages of training should be handled with care, so that the undertrained agents are not close to unreasonable behavior. In addition, exploration of the environment by the agent should be guided rather than completely random. Another important takeaway is to generally avoid using ensembles of agents as a single agent since they can incur unnatural behavior shifts, which can disorient the human player. Developing populations of diverse agents is still underexplored, even though having competent and diverse opponents during training is important for improving performance, especially when applying self-play. For DDA, it is frequent to have the agent minimize the absolute difference in health between players, but it may produce unnatural behavior. Approaches using different objectives are still heavily unexplored, and care should be taken as the objectives themselves may not align with the player's expectations of an optimal game experience.

All works that attempted transfer learning corroborate its success in fighting games, both when considering transferability to playing as and against different characters.

There are still gaps in current research. The impact of some methods on achieving qualities of human-centered agents is yet to be studied, e.g., planning methods for generalizability.

VIII. Future Directions

In terms of environments, we can note that both DIAMBRA Arena and FightLadder have a collection of environments that are deserving of further study. Neither of them take reaction time into consideration, which is important for fighting games. The use of audio features for the development of agents is still underexplored, which is currently the focus of DareFightingICE.

In terms of performance, there is some nontransitivity between agent strategies in the literature. Performance is also hard to compare between approaches due to varied evaluation methodologies and opponents. As such, the research community can benefit from tournament-based benchmarks featuring various opponents, such as the one in [119]. The opponent selection in such a benchmark can also be used for population-based training, which proves to be useful for achieving good performance [30]. FightLadder is one such benchmark yet to be explored, focused on multiagent RL [30], although the selection of opponents is not as large as in [119] (5 versus 43) and was only trained on one of the provided games. Still, the benchmark lays a framework for evaluation of agents through four criteria, which can help standardize how agents are evaluated in the literature and make performance comparisons easier.

To the best of the authors' knowledge, there is a lack of works exploring quick adaptation of agents to different opponents in fighting games. At least in the realm of RL, it might be worth exploring model-based methods for few-shot learning, by learning a model of the environment. Learning an environment model may be alleviated since we can make use of the game itself as the environment model, such as the simulator in DareFightingICE, focusing merely on learning the opponent model. Model-free methods have the disadvantage of learning exclusively through experience; as such, the agent cannot learn from hypothetical situations. This contrasts with the way human players learn these games, which is by speculation. Fighting games usually provide modes for aiding learning. One is a "training mode" where players can manually set up specific gameplay scenarios to learn from them, without having to actually experience them against other players. Another is a "replay mode," where players can scrutinize past recorded situations to understand them better, and possibly even interact with them to see what would happen had each player acted differently. We may take inspiration from this, exploring model-based methods for efficient learning by focusing on uncertain or rarely experienced situations.

There is no standard method to measure adaptability. One possible way to determine adaptability is to measure how many time steps it takes for the agent to achieve a neutral result against the current opponent. This neutral result can be, for example, 50% win rate or an average health difference of zero as in [13]. Since the duration of time steps is standardized in these games, we can have a reliable measure of how long it takes an agent to adapt in real time. The metric being evaluated, either win rate or average health difference, requires sampling multiple games or time steps to be accurate, so a minimum amount of games or time steps needs to be enforced.

For RL, we can explore maximum entropy RL to build robust agents that can more reliably face arbitrary opponent strategies [120]. Robustness may therefore help mitigate the nontransitivity in agent strategies. In addition, we can attempt to use intrinsic reward formulations, such as "curiosity"[87], [121] to guide exploration and facilitate online learning of countermeasures against unexpected opponent strategies.

For human likeness, all works that addressed reaction time considered a constant (visual) reaction time. This is not entirely realistic, however, since it assumes *simple* reaction time. Rather, the prevalent form of reaction time in these games is *choice* reaction time. That is, instead of the player having to react to a single stimulus, it needs to react to multiple stimuli, with each one requiring different responses. The different stimuli constitute the different kinds of moves that the opponent can perform, and the player's expectations of which will be performed influence reaction time, as these games are heavily based on prediction. This means that the more uncertain the player is of the future, the greater the reaction time is. Ideally, the choice reaction time is the one that should be emulated, or else we are strictly dividing situations between those that a player can reliably react to and those that they cannot. In these games, there is nuance between actions that are easy and those that are hard to react to, rather than a hard threshold between those that can always be reacted to or not. Also, it can be argued that a 15-frame reaction time, corresponding to 250 ms and the one most commonly emulated,

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这两种方法甚至也是商业游戏中最常选用的方法。另一方面，强化学习往往会产生人工行为，即使将其与模仿学习相结合也是如此。尽管强化学习取得的成功较少，但我们可以看到，分层强化学习和策略梯度方法产生的行为，比该范式中的其他方法（如基于价值的方法）更具类人性。然而，许多关于类人性的研究并未考虑智能体的公平性，例如对反应时间的仿真。反应时间约束对类人行为的影响仍有待确定。

对于如何开发具有吸引力的人工智能体，也有一些关键启示。首先，通过处于不同训练阶段的智能体版本来实现不同难度等级时，应谨慎处理，以免训练不足的智能体表现出近乎不合理的行为。此外，应引导智能体对环境的探索，而非使其完全随机进行。另一个重要启示是，通常应避免将智能体集成作为单个智能体使用，因为这可能导致不自然的行为转变，使人类玩家感到困惑。对多样化智能体种群开发的探索仍然不足，尽管在训练过程中拥有能力出众且多样化的对手对于提升性能十分重要，尤其是在采用自博弈时。在动态难度调整中，常见做法是让智能体尽量缩小玩家之间生命值的绝对差，但这可能产生不自然的行为。对于采用其他目标的方法，目前的探索仍严重不足，而且需要谨慎对待，因为这些目标本身可能与玩家对最佳游戏体验的期望不一致。

所有尝试迁移学习的研究均证实了其在格斗游戏中的成功，无论是迁移到操控不同角色，还是迁移到对抗不同角色。

当前研究仍存在空白。一些方法对于实现以人为中心的智能体所需特性的影响尚待研究，例如规划方法对泛化能力的影响。

VIII. 未来方向

在环境方面，我们注意到，DIAMBRA Arena 和 FightLadder 都提供了一系列值得进一步研究的环境。二者均未考虑反应时间，而这对格斗游戏至关重要。利用音频特征开发智能体的研究仍不充分，而这正是 DareFightingICE 目前关注的重点。

在性能方面，文献中的智能体策略之间存在一定的非传递性。由于评估方法和对手各不相同，不同方法之间的性能也难以比较。因此，研究界可以受益于包含多种对手的基于锦标赛的基准，例如 [119]中的基准。此类基准中的对手选择也可用于基于种群的训练，事实证明，这有助于取得良好的性能 [30]。FightLadder 就是这样一个尚待探索的基准，专注于多智能体强化学习 [30]，不过其可选对手的数量不如 [119] 中的多（5 个对 43 个），而且仅在所提供的其中一款游戏上进行了训练。尽管如此，该基准仍建立了一个从四项标准评估智能体的框架，有助于规范文献中的智能体评估方式，并使性能比较更加容易。

据作者所知，目前缺乏探索格斗游戏中智能体如何快速适应不同对手的研究。至少在强化学习领域，通过学习环境模型来实现小样本学习的基于模型的方法可能值得探索。由于可以将游戏本身用作环境模型，例如 DareFightingICE 中的模拟器，因此可以减轻学习环境模型的负担，仅专注于学习对手模型。无模型方法的缺点是只能通过经验学习，因此智能体无法从假设情境中学习。这与人类玩家通过推想来学习这些游戏的方式形成了对比。格斗游戏通常提供辅助学习的模式。一种是“训练模式”，玩家可以手动设置特定的游戏场景并从中学习，而不必在与其他玩家对战时实际经历这些场景。另一种是“回放模式”，玩家可以仔细审视过去录制的情境，以便更好地理解它们，甚至可能与这些情境互动，看看如果每位玩家采取不同的行动，会发生什么。我们可以从中获得启发，探索基于模型的方法，通过聚焦于不确定或很少经历的情境来实现高效学习。

目前尚无衡量适应性的标准方法。一种可行的适应性评估方式，是衡量智能体需要多少个时间步才能在对抗当前对手时达到均势。这种均势可以是例如 50% 的胜率，或如 [13]中那样，平均生命值差为零。由于这些游戏中时间步的时长是标准化的，因此我们可以可靠地衡量智能体适应所需的实际时间。所评估的指标，无论是胜率还是平均生命值差，都需要对多场游戏或多个时间步进行采样才能确保准确，因此必须规定最少的游戏场数或时间步数。

对于强化学习，我们可以探索最大熵强化学习，以构建能够更可靠地应对任意对手策略的鲁棒智能体 [120]。因此，鲁棒性可能有助于缓解智能体策略中的非传递性。此外，我们可以尝试使用“好奇心”等内在奖励构造[87]，[121] 来引导探索，并促进针对意料之外的对手策略进行应对策略的在线学习。

在类人性方面，所有涉及反应时间的研究都采用了恒定的（视觉）反应时间。然而，这并不完全符合实际，因为它假定的是简单反应时间。实际上，这类游戏中更常见的反应时间形式是选择反应时。也就是说，玩家并非只需对单一刺激作出反应，而是需要对多种刺激作出反应，且每种刺激都要求不同的应对方式。这些不同的刺激就是对手可以施展的各类招式，而玩家对对手将施展何种招式的预期会影响反应时间，因为这类游戏高度依赖预测。这意味着，玩家对未来越不确定，反应时间就越长。理想情况下，应当模拟选择反应时，否则就会将情境严格划分为玩家能够可靠作出反应的情境和无法作出反应的情境。在这类游戏中，容易应对的动作与难以应对的动作之间存在细微差别，而非一道将始终能够作出反应与无法作出反应截然分开的界限。此外，也可以认为，最常被模拟的15帧反应时间（相当于250毫秒），

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is difficult to achieve by human players in practice. As such, current works are optimistic in the human abilities assumed for human-like artificial agents. However, it is not clear how choice reaction time can be emulated. There are different models for choice reaction time drawn from experimental data [49], [122], [123], but it is uncertain which model should be used. Since fighting games require sequential decision making, the work in [122] might be the most appropriate. Still, even with a model defined it is not obvious how reaction time should be modulated. It is possible to use uncertainty in an opponent model or environment model, or perhaps the breadth of a search tree in short-term planning, and even after reaction time is properly emulated, we have the issue of handling a variable reaction time, which has already been explored for RL [124], [125], [126]. This is a hard challenge when creating human-like agents using methods other than imitation-based ones, such as RL.

If we want to achieve fairness, then we need to address the fact that visual processing as done by human players affects decision making. It is possible to use visual ambiguity and occlusion as part of one's strategy, which affect the opponent's perception of the game state. When using structured representations of the game state, we are thus losing some fairness in this regard. Still, as long as the game has an appropriate sound design, these games may also be played through sound alone, so this factor loses some weight. Furthermore, parsing multidimensional input is usually done using convolutional neural networks, which are computationally expensive when applied to the ordinary hardware that players have access to. The importance of this factor can be debated, but we believe it is still worth mentioning.

For difficulty adjustment, we can explore creating agents for multiple difficulties by limiting their action space instead [52], which reduces the complexity of the game and makes it less overwhelming for newer players. Another possibility is to have the agent only execute the desired actions with a given probability, emulating the action execution difficulty of these games [52]. To determine player skill in DDA, we can make use of different metrics, such as those in [127]. Training multiple performant agents for different action space sizes or action execution probabilities is an interesting avenue for future work. We can also try to create easier agents by increasing the predictability of the agent's behavior, e.g., by lowering the temperature when using the softmax function for building action probability distributions.

Evolutionary algorithms are a great candidate for building large pools of diverse agents since these are a natural artifact of the algorithms [114], but they have still not been explored extensively. Quality-diversity algorithms are a class of evolutionary algorithms that aims to create populations of individuals that are not only fit but also diverse [128], and are yet to be explored in fighting games.

In general, there is yet to be a unified method that fulfills all qualities of a human-centered agent. We may combine the strengths of the methods presented in this work. For instance, we can employ SL to achieve human likeness, and RL to improve the agent's performance. Planning can be further incorporated to improve the agent's decisions, and evolutionary algorithms can be explored to create diverse agents. Finally, we may perform some manual instrumentation on the agent, as rule-based methods provide predictability and allow handling exceptional cases, at the cost of generality.

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在人类玩家的实际操作中很难达到。因此，当前研究对类人人工智能体所假定的人类能力较为乐观。然而，如何模拟选择反应时尚不明确。已有多种根据实验数据建立的选择反应时模型 [49],[122], [123], ，但应当使用哪种模型仍不确定。由于格斗游戏需要序贯决策， [122] 中的研究可能最为适用。不过，即使确定了模型，如何调节反应时间也并不显而易见。可以利用对手模型或环境模型中的不确定性，也可以利用短期规划中搜索树的广度；而即使已经恰当地模拟了反应时间，我们仍需处理可变反应时的问题，强化学习领域已有研究对此进行了探索 [124], [125], [126]。在使用强化学习等非基于模仿的方法创建类人智能体时，这是一个艰巨的挑战。

如果我们希望实现公平性，就需要考虑人类玩家的视觉处理会影响决策这一事实。玩家可以将视觉歧义和遮挡作为自身策略的一部分，以影响对手对游戏状态的感知。因此，使用游戏状态的结构化表示会在这方面损失一定的公平性。不过，只要游戏具有适当的声音设计，玩家也可以仅凭声音进行游戏，因此这一因素的重要性会有所降低。此外，多维输入的解析通常使用卷积神经网络，而在玩家能够使用的普通硬件上运行这类网络时，计算开销较大。这一因素的重要性或许存在争议，但我们认为仍值得一提。

在难度调整方面，我们可以探索通过限制智能体的动作空间来创建具有不同难度的智能体 [52]，从而降低游戏的复杂性，使新手玩家不至于感到过于吃力。另一种可能的方法是让智能体仅以给定概率执行预期动作，以模拟这类游戏的动作执行难度 [52]。为了在动态难度调整中确定玩家技能，我们可以采用不同的指标，例如 [127]中的指标。针对不同的动作空间大小或动作执行概率，训练多个表现出色的智能体，是未来工作中一个值得探索的方向。我们还可以尝试提高智能体行为的可预测性，以创建更容易应对的智能体，例如在使用softmax函数构建动作概率分布时降低温度。

进化算法非常适合构建包含大量多样化智能体的集合，因为这些智能体是算法自然产生的结果， [114], 但这类算法尚未得到广泛探索。质量多样性算法是一类进化算法，旨在创建不仅适应度高而且具有多样性的个体种群， [128], 目前尚未在格斗游戏中得到探索。

总体而言，目前尚无一种统一的方法能够满足以人为中心的智能体的所有特质要求。我们可以结合本文所介绍的各种方法的优势。例如，可以采用监督学习来实现类人性，并利用强化学习来提升智能体的性能。还可以进一步融入规划，以改进智能体的决策，并探索利用进化算法来创建多样化的智能体。最后，我们可以对智能体进行一些人工干预，因为基于规则的方法能够提供可预测性，并允许处理特殊情况，但代价是牺牲通用性。

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